FINAL RECAP

Francesco Zonta, Ph.D.

Associate Professor

Department of Biological Sciences

Room SB465, francesco.zonta@xjtlu.edu.cn



Xi'an Jiaotong-Liverpool University

西交利物浦大學

Structure of the exam

4 open questions – closed book. Testing understanding of the theory.

Key concepts to know:

1) Basic knowledge about molecular physics.
2) Structure of biomolecules, DNA, RNA, proteins.
3) Force fields and interatomic potentials.
4) Docking and folding.
5) Practical skills (understanding files structure, understanding basic algorithms, molecular screening etc.).

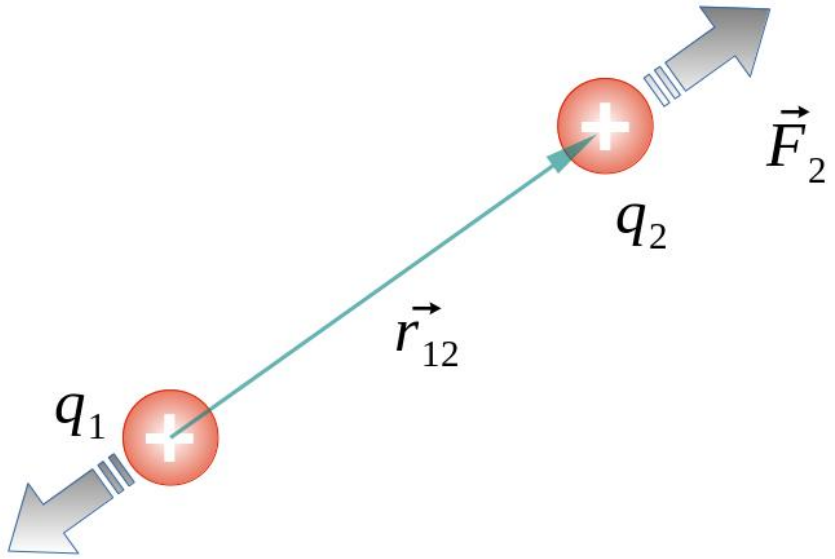
Coulomb law



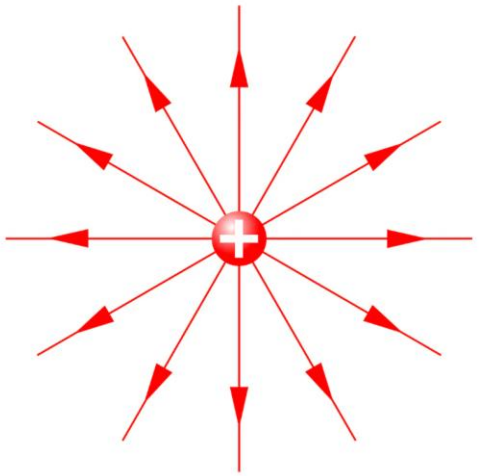
Forces are vectors: the force between two charges in a 3d space has the same direction of the line connecting the two charges:

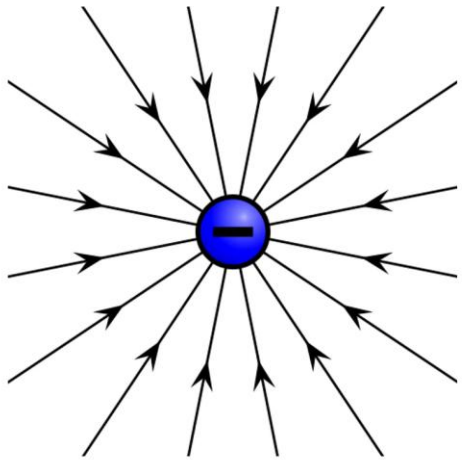
$$\boldsymbol{F} = k\frac{q_1 q_2}{r_{12}^2}\boldsymbol{u}_{r_{12}} = k\frac{q_1 q_2}{r_{12}^3}\boldsymbol{r}_{12}$$

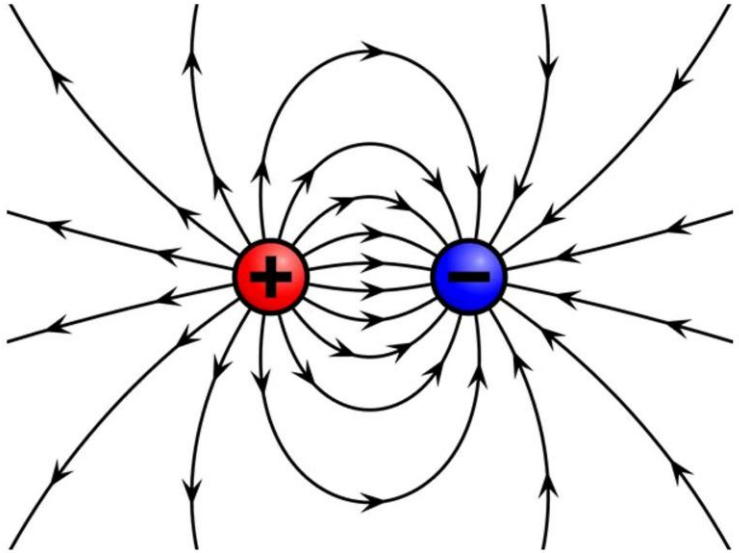
$\boldsymbol{u}_{r_{12}}$ is a unitary vector in the direction of $\boldsymbol{r}_{12}$



The electric field







$$\boldsymbol{F} = q \cdot k \frac{Q}{r_{12}^2} \boldsymbol{u}_{r_{12}}$$

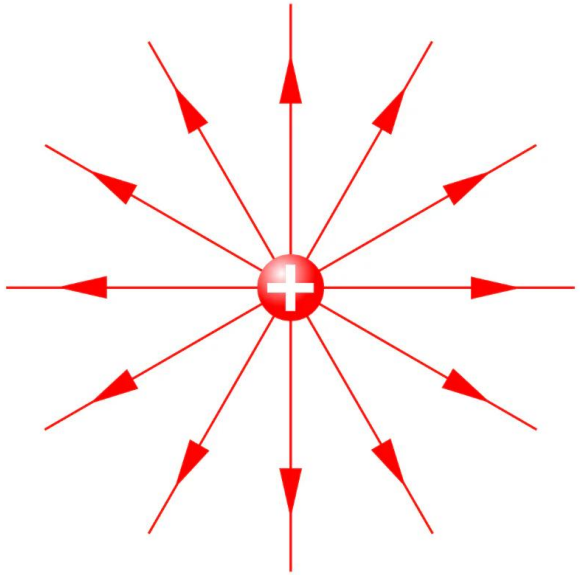
Suppose we keep a charge q fixed in a point in space, and we measure the force $\boldsymbol{F}$ that a second charge q feels, as it moves around the charge Q.

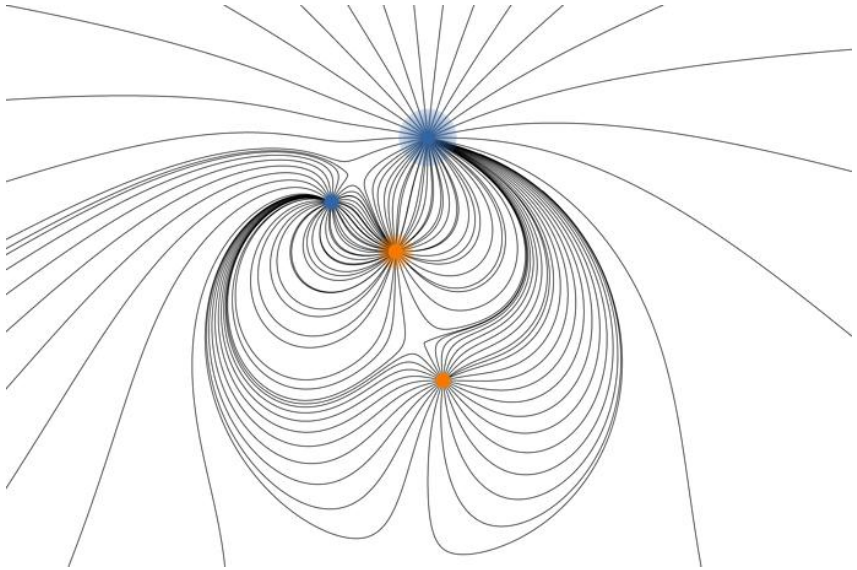
Since the force $\boldsymbol{F}$ is proportional to q, the ratio between $\boldsymbol{F}$ and q_2 depends only on the specific position in space:

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{\boldsymbol{F}(\boldsymbol{r})}{q}$$

We call $\boldsymbol{E}(\boldsymbol{r})$ the electric field generated by Q and we can think at it as a perturbation of the empty space, due to the existence of Q.

The electrostatic potential





The potential energy of a charge q, which is interacting with another charge Q is proportional to q.

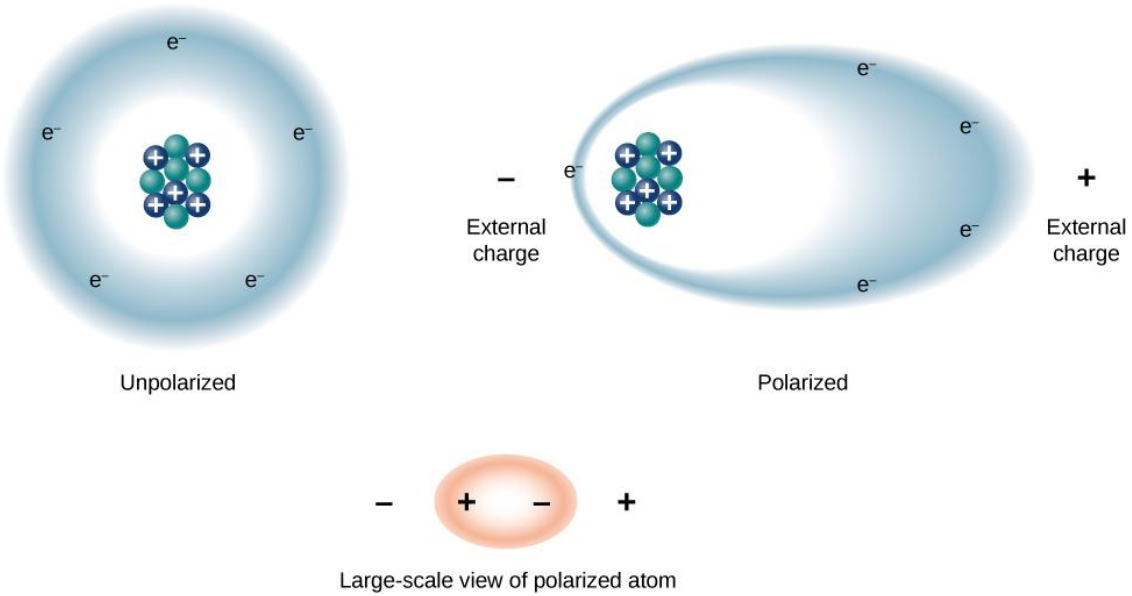
$$\boldsymbol{F} = k\frac{qQ}{r_{12}^2}\boldsymbol{u}_{r_{12}} \Rightarrow U = +k\frac{qQ}{r_{12}} = q\left(+k\frac{Q}{r_{12}}\right)$$

This is true in general, the potential energy felt by a charge q is proportional to the charge itself (even if in general the field can be generated by many charges and have a very complicated form).

Similarly to the definition of the electric field, we can now define the electrostatic potential as the ration between the electrostatic potential energy and the charge that feels that energy:

$$V(\boldsymbol{r}) = \frac{U(\boldsymbol{r})}{q} \qquad \boldsymbol{E} = -\boldsymbol{\nabla V}$$

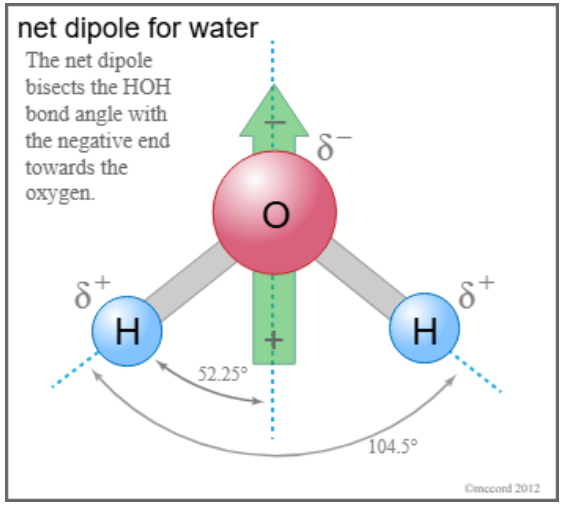
Polarization of atoms and molecules



Electrons are distributed in orbitals around the atomic nuclei.

When an external electrostatic field is applied, the electron cloud shifts slightly relative to the nucleus.

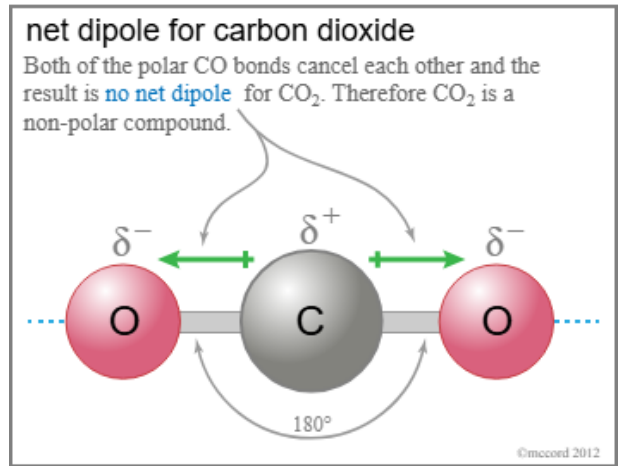
This creates an electric dipole moment which tends to align with the external field.



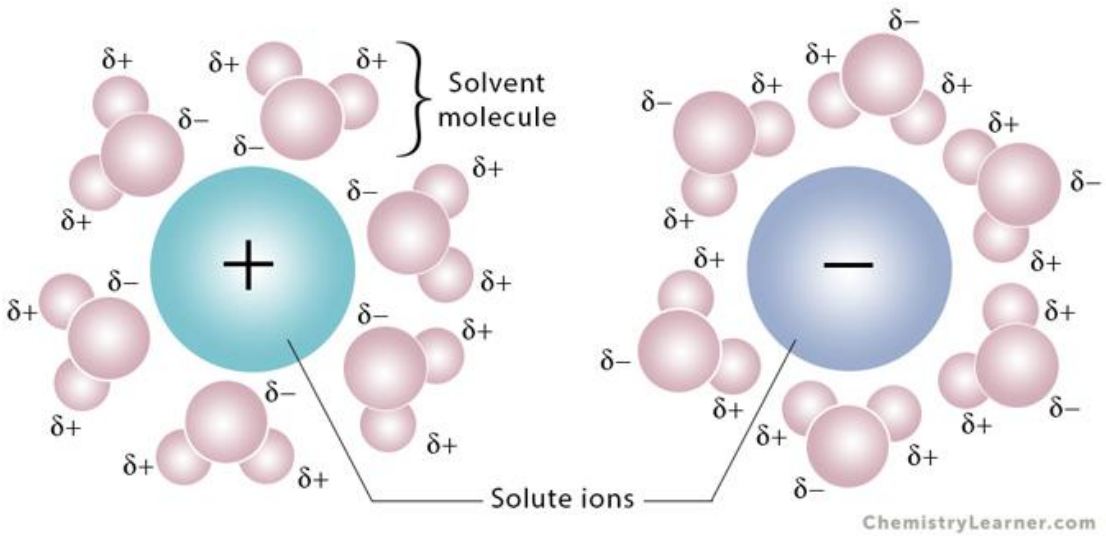
The electrostatic attraction between electrons and nuclei differs for different atoms.

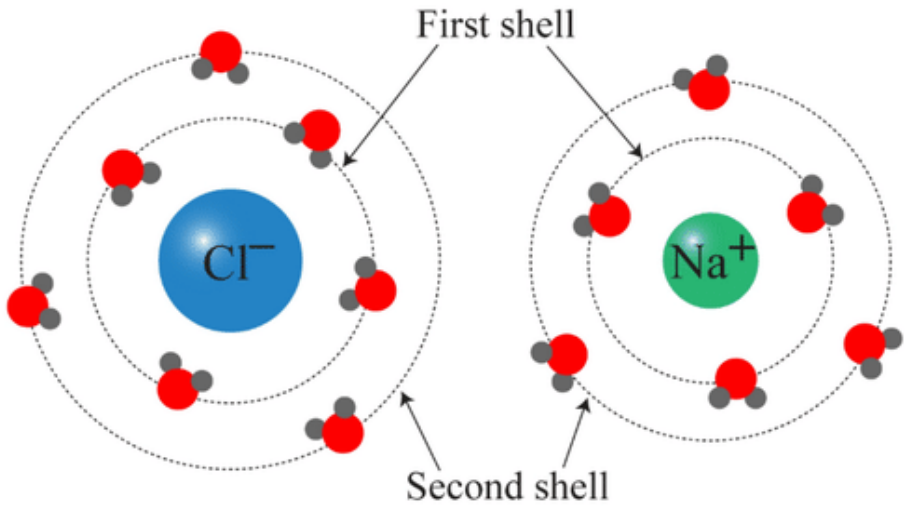
As a result, in a molecule, electrons may be more localized around one atom than another, leading to partial charges.

Depending on the geometry, some molecules have a permanent dipole moment, even in the absence of external fields.



Solvation (hydration) shells





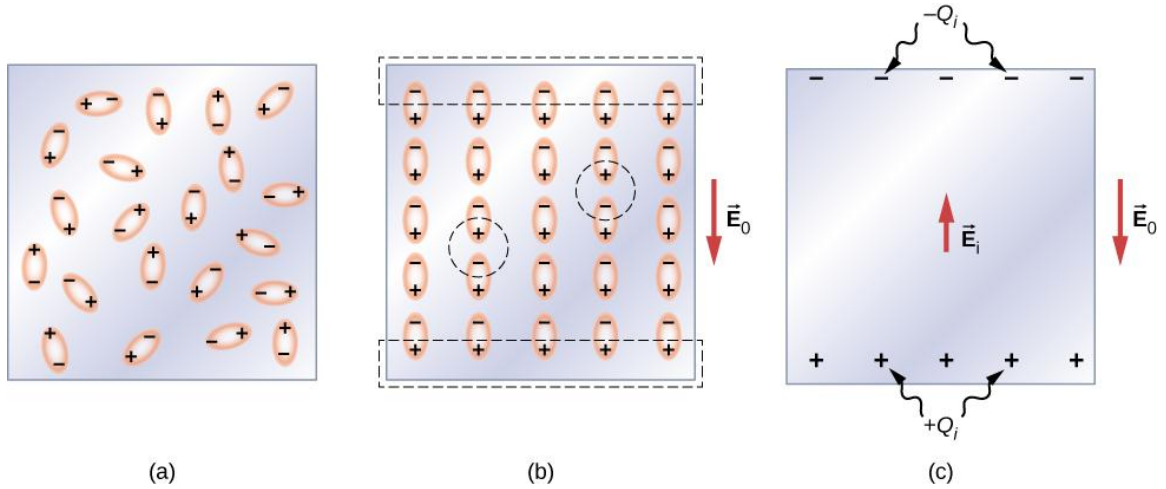
A solvation shell is the structured layer of solvent molecules surrounding a solute (e.g., an ion).

Because water is polar and ions are charged, solvation shells are relatively stable and can mask the ion's charge.

Multiple solvation shells may form, but they become progressively less ordered and less stable farther from the ion.

Solvation shells are dynamic, with solvent molecules constantly exchanging between the shells and the bulk.

Water dielectric constant



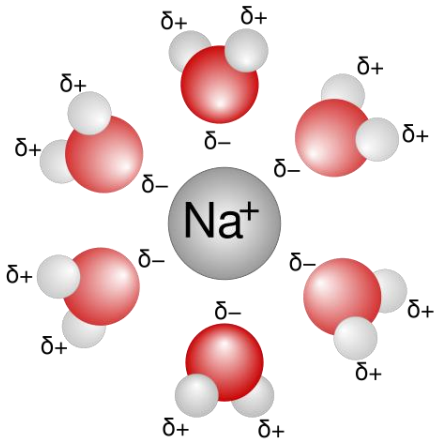
When an external electric field is applied, molecular dipoles (like water) tend to reorient in the direction of the field.

This collective reorientation creates an induced field that opposes the external field, partially canceling it.

The stronger the dipoles and their alignment, the greater the reduction of the effective field inside the medium.

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \rightarrow E = \frac{1}{4\pi\varepsilon_0\varepsilon_r}\frac{q}{r^2}$$

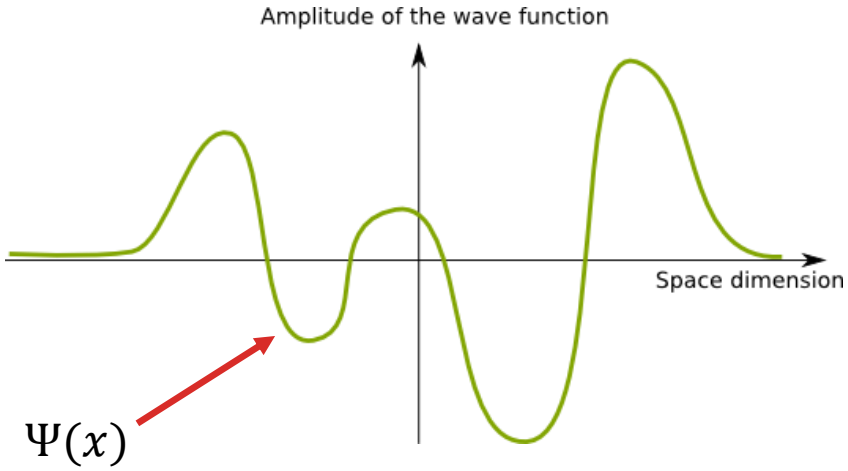
In vacuum In water



The dielectric constant (ε_r) quantifies this effect: it measures how much a medium reduces the effective electrostatic interaction between charges compared to vacuum.

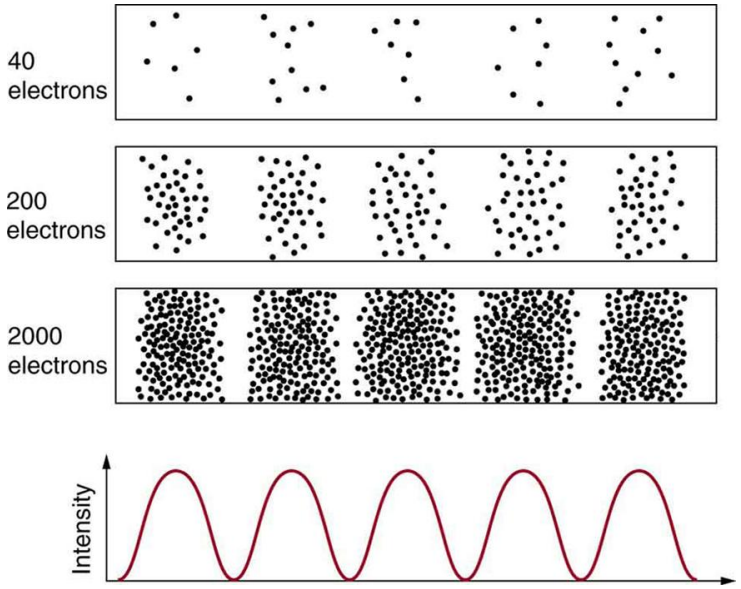
Water has a high dielectric constant (~80 at room temperature), meaning it strongly screens charges and stabilizes ions and polar molecules in solution.

Wave functions



To describe this behavior, we introduce the mathematical concept of wave function.

$\Psi(x)$: wave function.
It represent the quantum state of the system



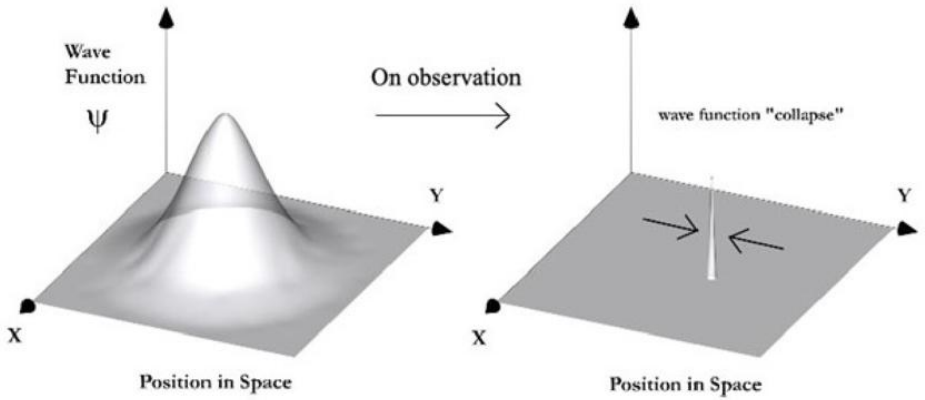
The function $\Psi(x)$ can be used to determine the probability that a particle (i.e. an electron, e) is found in a certain part of the space (i.e the interval [a,b]).

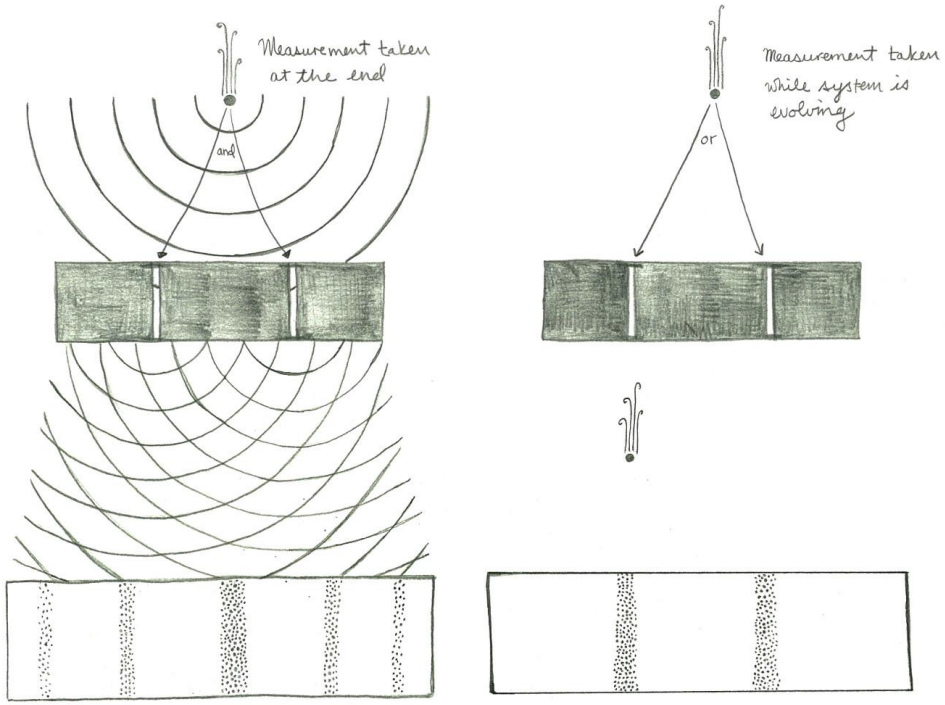
$$P(e \in [a,b]) = \int_a^b |\Psi(x)|^2 dx$$

Mathematically, the wave function is a complex function.
This is way we take the modulus squared of $\Psi(x)$.

Francesco Zonta – Room SB465, Department of Biosciences and Bioinformatics – francesco.zonta@xjtlu.edu.cn

Measurements and collapse of the wave function





When we take a measurement (for example the position of an electron), we do not get the wave function, but we get the result of the measurement (for example the coordinates x,y,z).

After the measurement, we know where the electron is, and it cannot be in all the other possible places.

A possible interpretation is that the measurement collapses the wave function!

The wave function changes due to the measurements.

The equation of Schroedinger

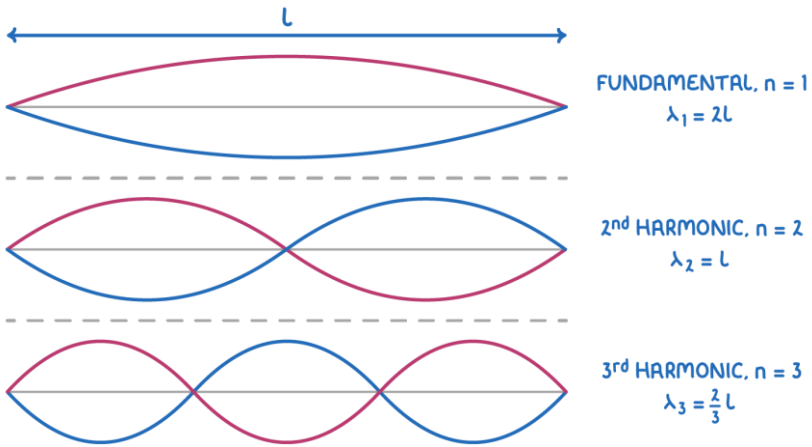


This equation describes the time evolution of the wave function:

$$i\hbar \frac{\partial}{\partial t}\Psi(x,t) = H\Psi(x,t)$$

$$i\hbar \frac{\partial}{\partial t}\Psi(x,t) = \left[\frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x,t)\right]\Psi(x,t)$$

Total energy of the system: $\mathrm{H} = \frac{1}{2}mv^2 + V = \frac{p^2}{2m} + V$

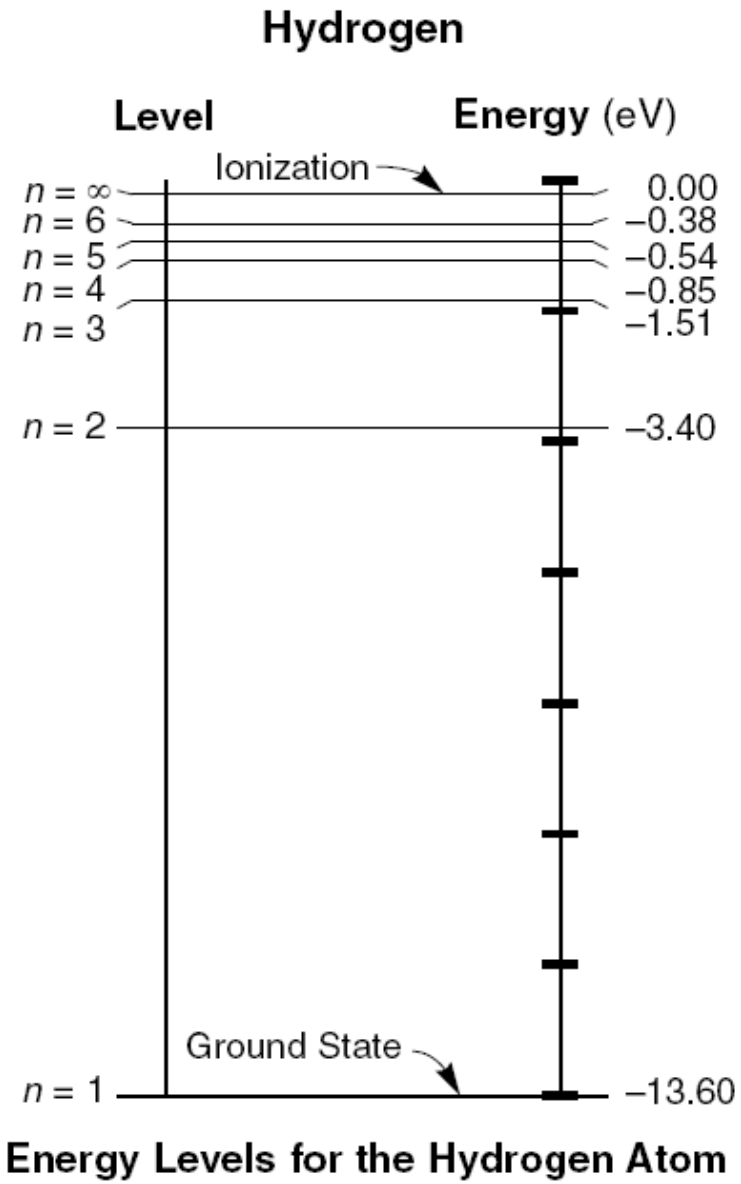


Harmonics are stationary patterns

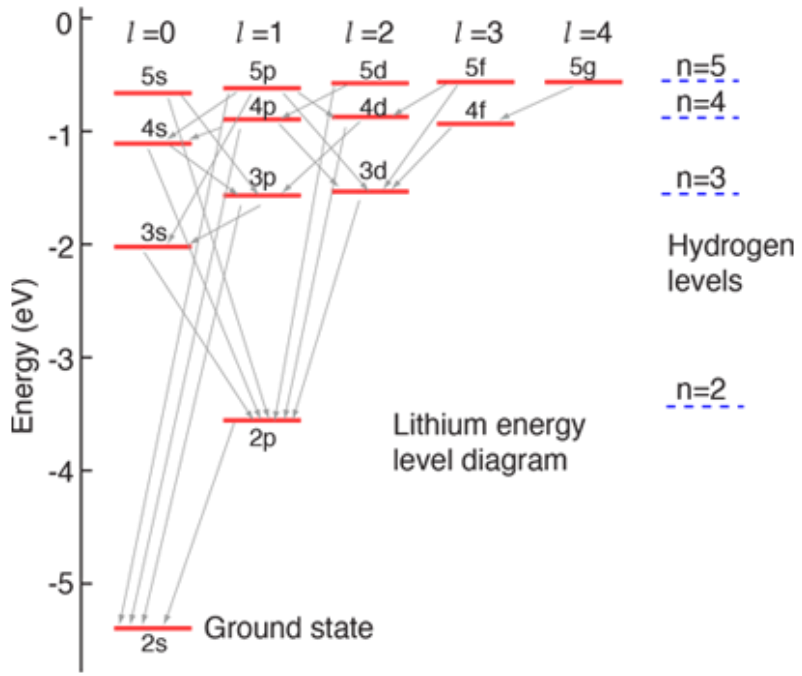
Time independent solutions of the equation of Shroedinger will obey the following property:

$$H\Psi(x) = E\Psi(x)$$

Orbitals



Energy Levels for the Hydrogen Atom



$$HR(r)Y(\theta,\phi) = ER(r)Y(\theta,\phi)$$

The equation of Shroedinger has solution only for discrete values of E.

The energy of an electron bound to an atom is quantized.

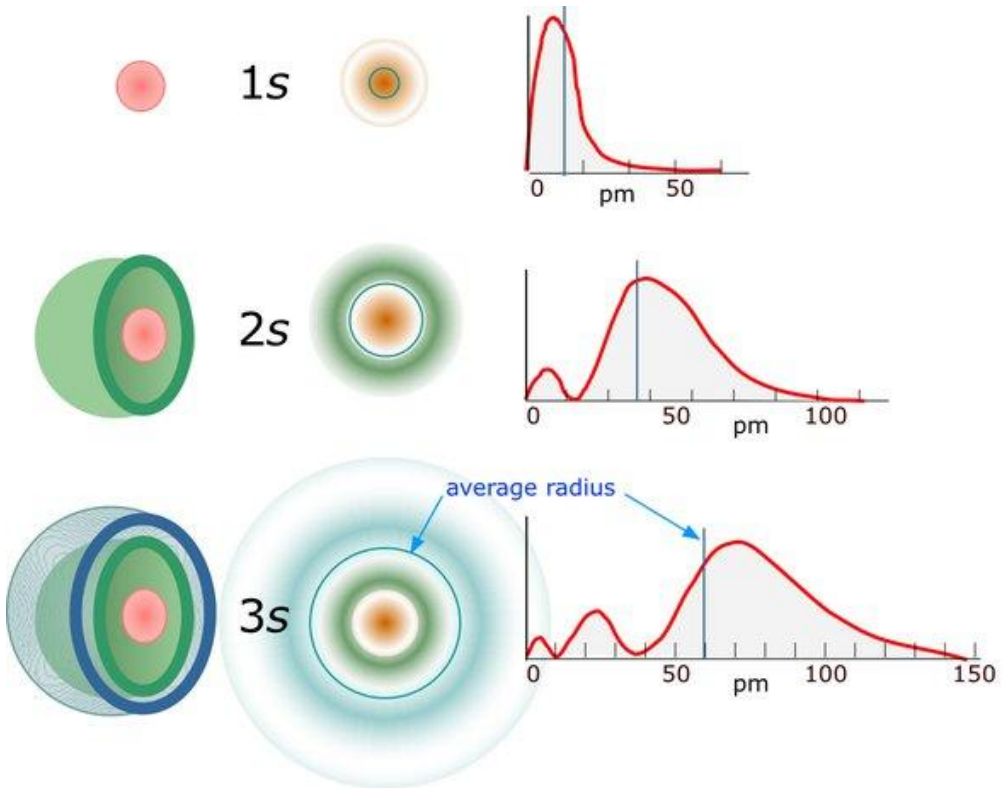
The solutions give us the energy values, orbital "sizes" and orbital "shapes".

For the hydrogen atom, the energy levels do not depend on the angular part of the solution.

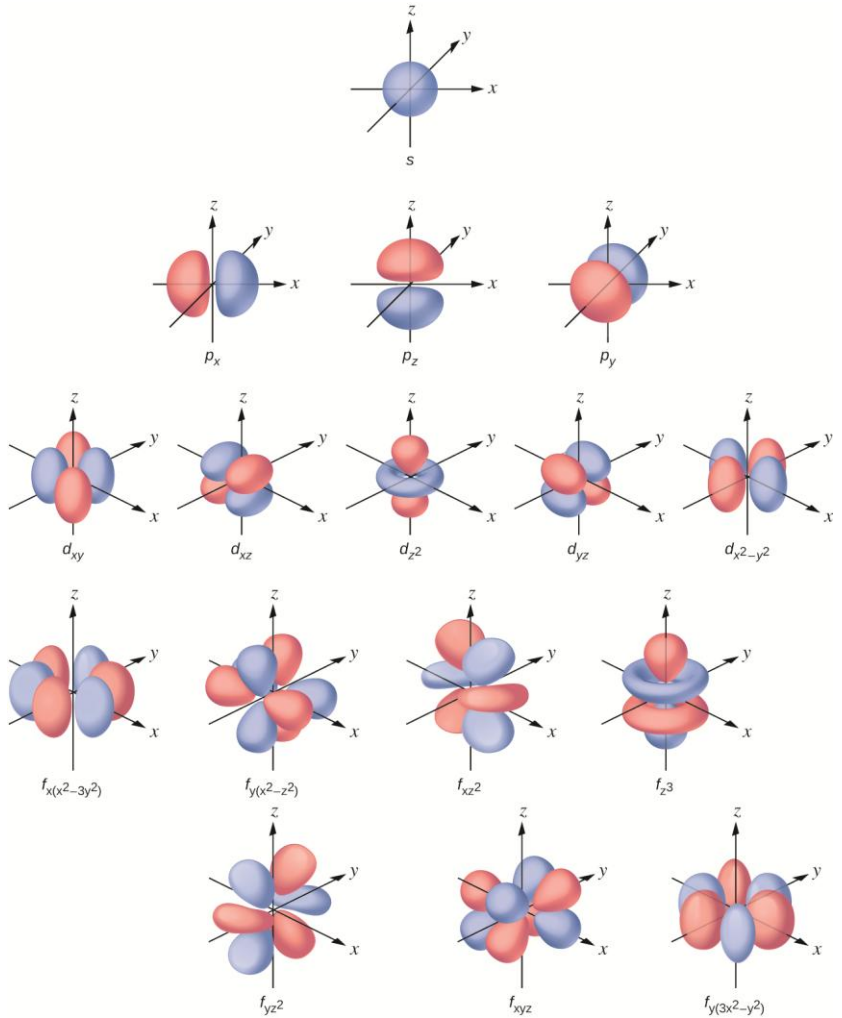
But in general different orbital shapes have different energy.

Orbitals

Orbitals are characterized by few quantum numbers that corresponds to the possible solutions of the Schroedinger equation for that atom.

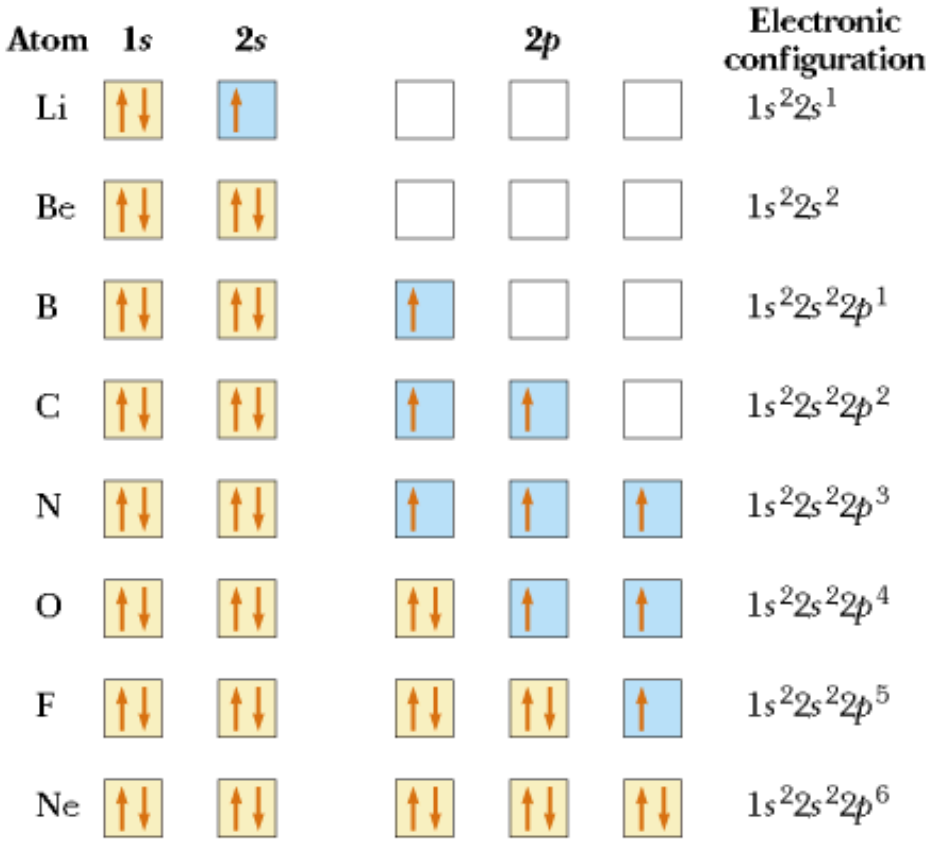


Radial part of the orbital



Angular part of the orbital

Spins and Pauli's exclusion principle



Serway, Physics for Scientists and Engineers, 5/e
Figure 42.14 Harcourt, Inc.

In addition to the quantum numbers that describe an orbital's energy and shape, electrons have an intrinsic quantum property called spin.

This property is quantized and can have one of two values: spin-up (+ 1/2) or spin-down (– 1/2).

Pauli's exclusion principle states that no two electrons in the same atom can have the same set of four quantum numbers.

An atomic or molecular orbital can hold a maximum of two electrons, and these two electrons must have opposite spins.

This explains why electron shells fill up in a specific order, which in turn determines the chemical properties of atoms and the nature of chemical bonds that form molecules.

Molecular orbitals

Energy for excitation: 10 eV

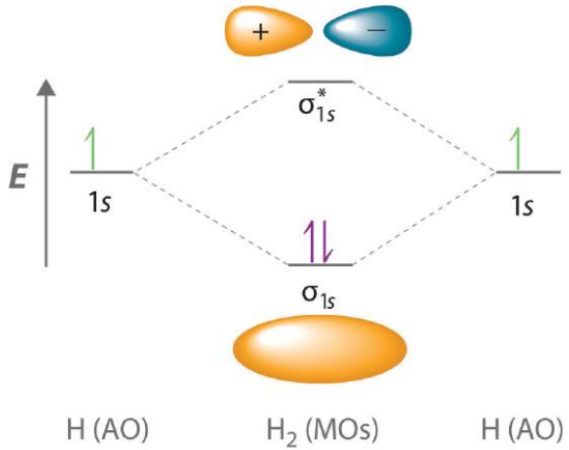
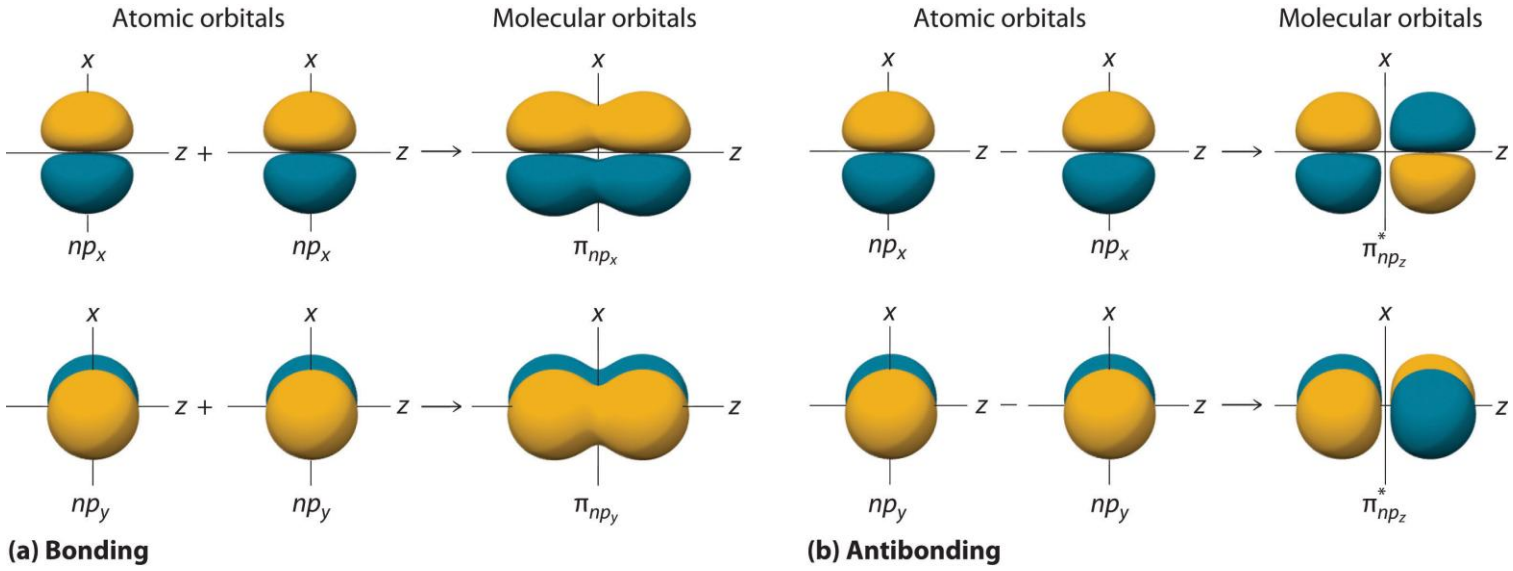
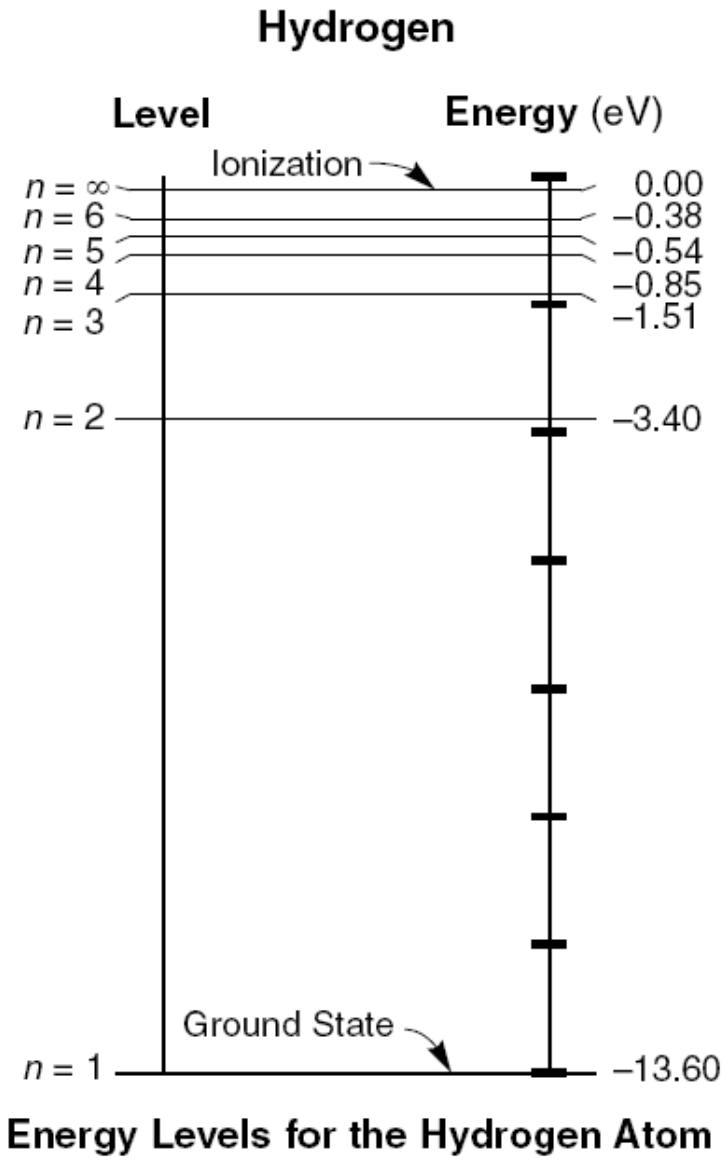


Figure 5.4.1.2: Molecular Orbital Energy-Level Diagram for H_2. The two available electrons (one from each H atom) in this diagram fill the bonding σ_{1s} molecular orbital. Because the energy of the σ_{1s} molecular orbital is lower than that of the two H 1*s* atomic orbitals, the H_2 molecule is more stable (at a lower energy) than the two isolated H atoms.



Classical Physics still works!



Energy Levels for the Hydrogen Atom

Fortunately, we live at 310K, and biology uses only few *light* atoms

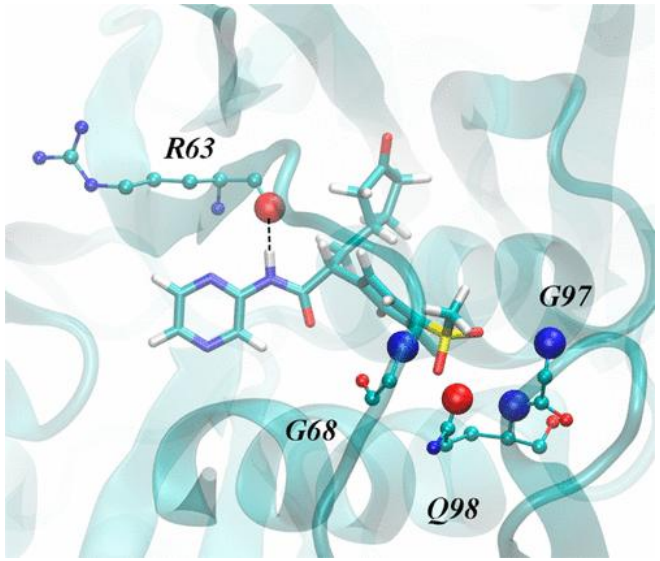
Thermal energy at room temperature: **26 meV**
Electronic energy levels: around **10 eV**
Probability of transition for thermal fluctuations: **10^{-171}**

The energy cost to excite light atoms is much higher than the thermal energy.

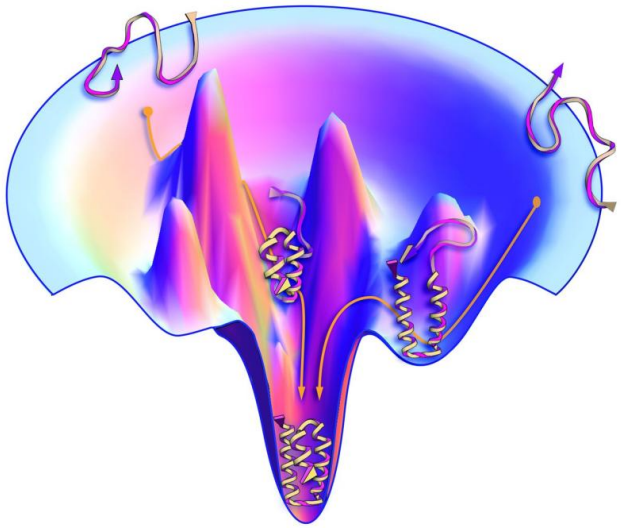
Light atoms (and molecules made of them) are all in the "ground state" at room temperature.

Chemical reactions and heavy atoms are very well controlled within any living organism.

Molecules according to classical physics



Mollica et al. J. Med. Chem. 2016



Each atom is treated as a "classical" object interacting with the other atoms in the system.

The interaction is represented through the classical potential energy.

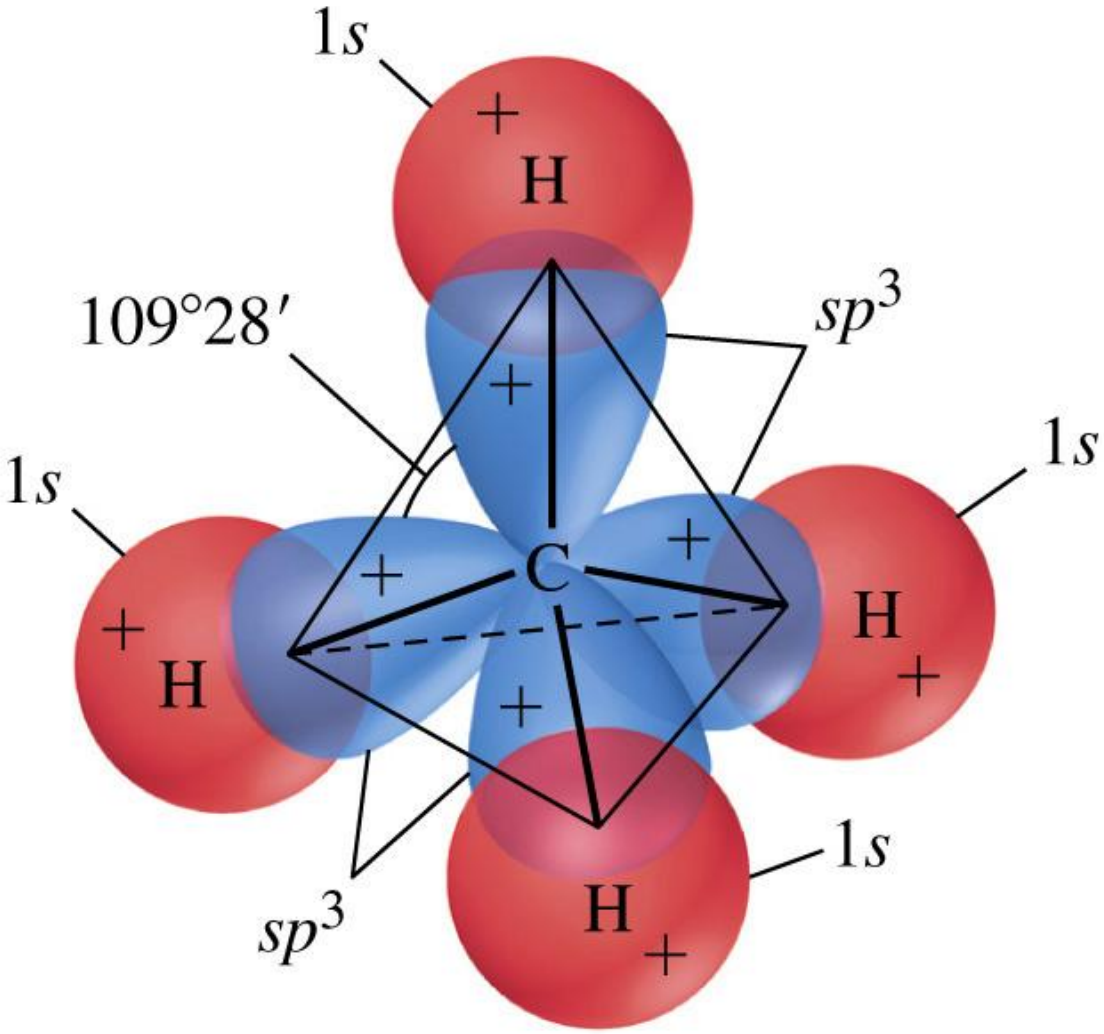
The equilibrium state is the minimum of the potential energy.

Different configurations are possible, without altering the electronic configurations of the molecules.

From the energy we can get the forces, and the time evolution.

At 310K different states are accessible, so Entropy exists, and Free Energy will decide the most probable state for the system.

Classical Force Fields



Since the electronic configuration is frozen in the ground state, we can simplify the form of the energy of interaction in many different ways:

1) A charge is assigned to each atom, and it is equal to the average charge of that atom.

2) Interactions between bonded atoms reproduce chemical and geometrical properties of the molecule, in particular regarding bond distances, angles and strength.

3) Non-bonded atoms interact exclusively through Coulomb and Lennard-Jones potentials.

Analytical form of the Potential Energy

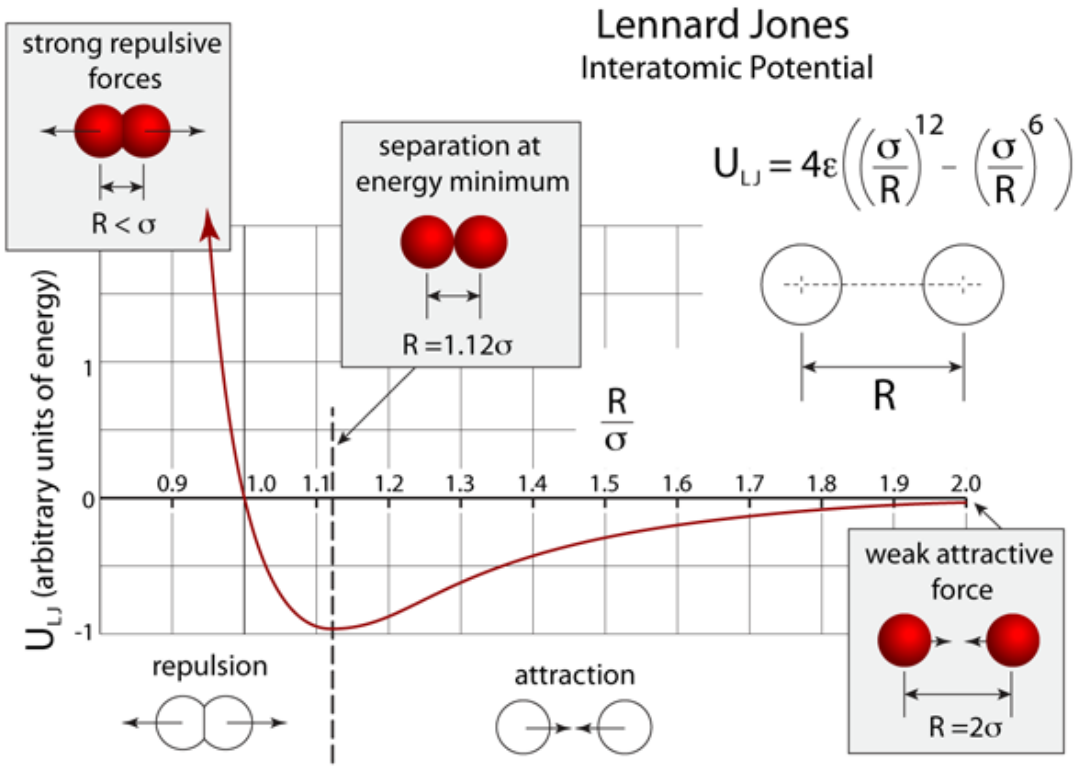
$$U = \sum_{bonds} k_r(r - r_0)^2 + \sum_{angles} k_\theta(\theta - \theta_0)^2 +$$

$$\sum_{dihedrals} C_k \cos(\varphi - \varphi_{0,k})) + \sum_{impropers} k_\alpha(\alpha - \alpha_0)^2 +$$

$$\sum_{i<j} \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}} + \sum_{i<j} \left(\frac{A}{r_{ij}^{12}} - \frac{B}{r_{ij}^{6}} \right) + \sum_{h.b.} \left(\frac{C}{r_{ij}^{12}} - \frac{D}{r_{ij}^{10}} \right)$$

This is one example of a classical force fields. Other possibilities exist.

The Lennard-Jones potential



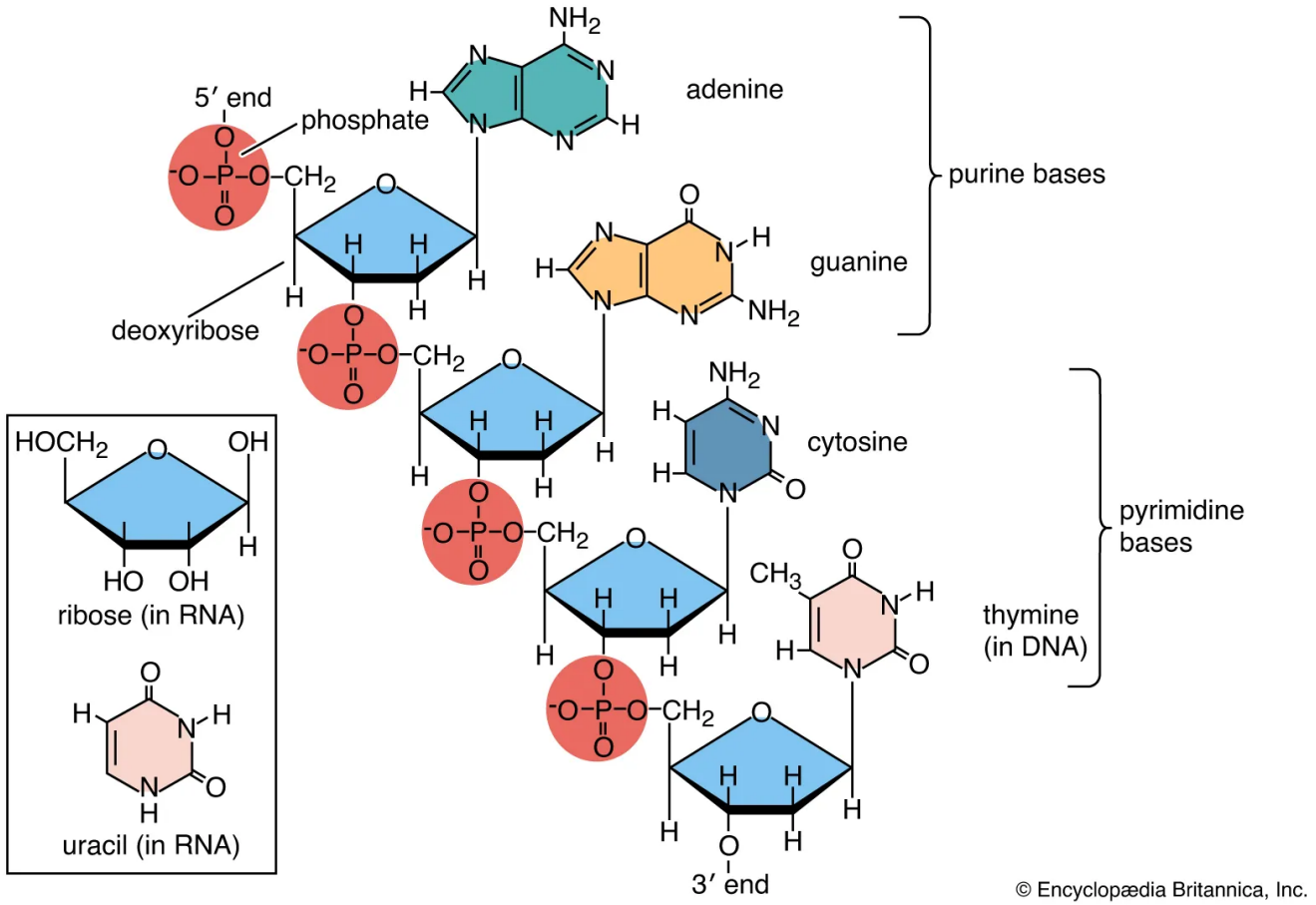
physicsatmcl.commons.msu.edu

At short distances, the LJ potential is strongly repulsive. Atoms that are too nearby in space create strong repulsions, that could be enough to completely distort the protein structure.

This happens very often when hydrogen atoms are added to the system (for example for a structure downloaded from the PDB or after a docking).

Energy has to be carefully minimized before starting MD.

Polymerization and pairing of nucleotides



© Encyclopædia Britannica, Inc.

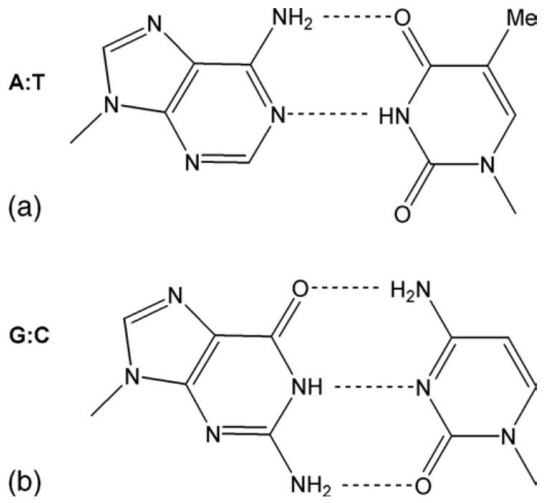
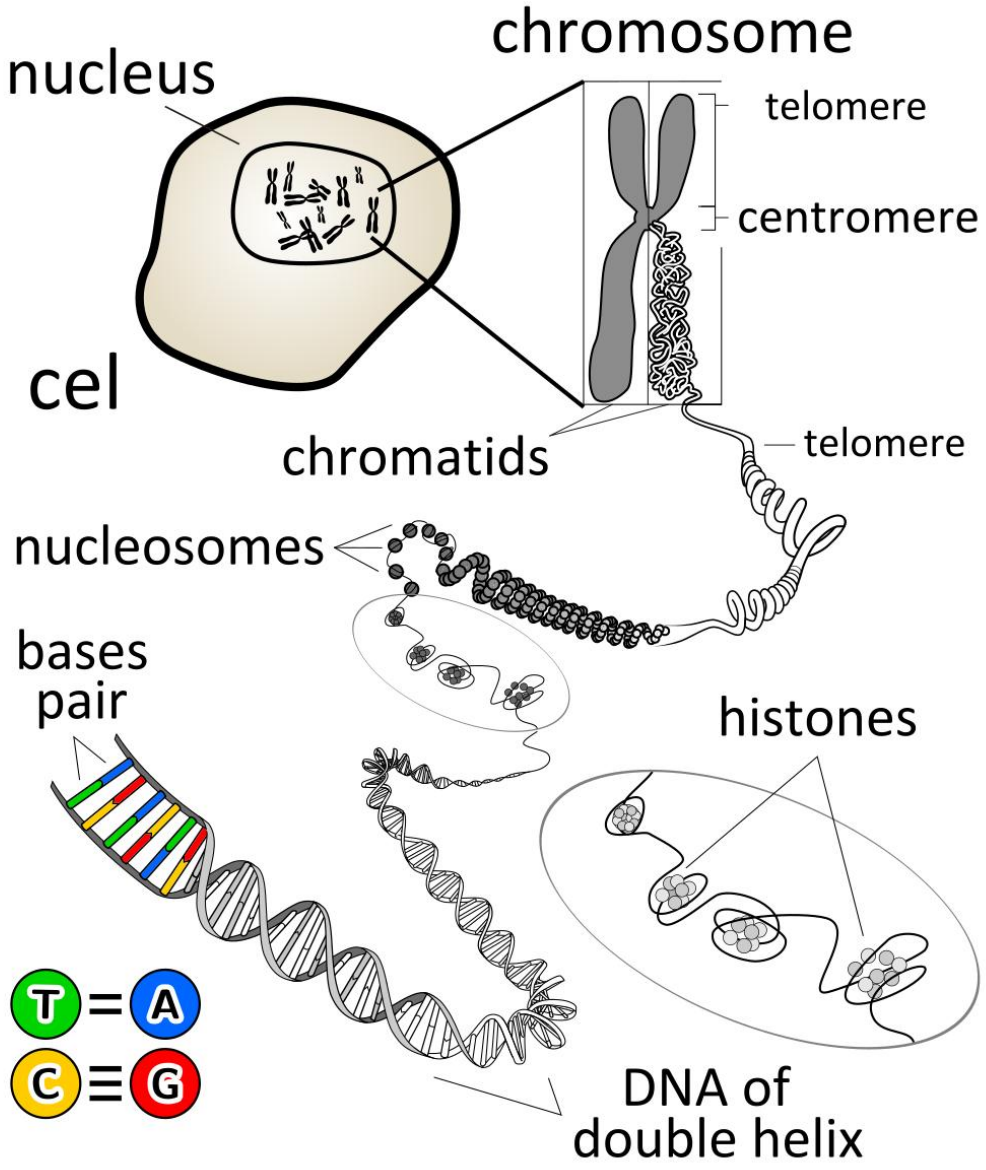


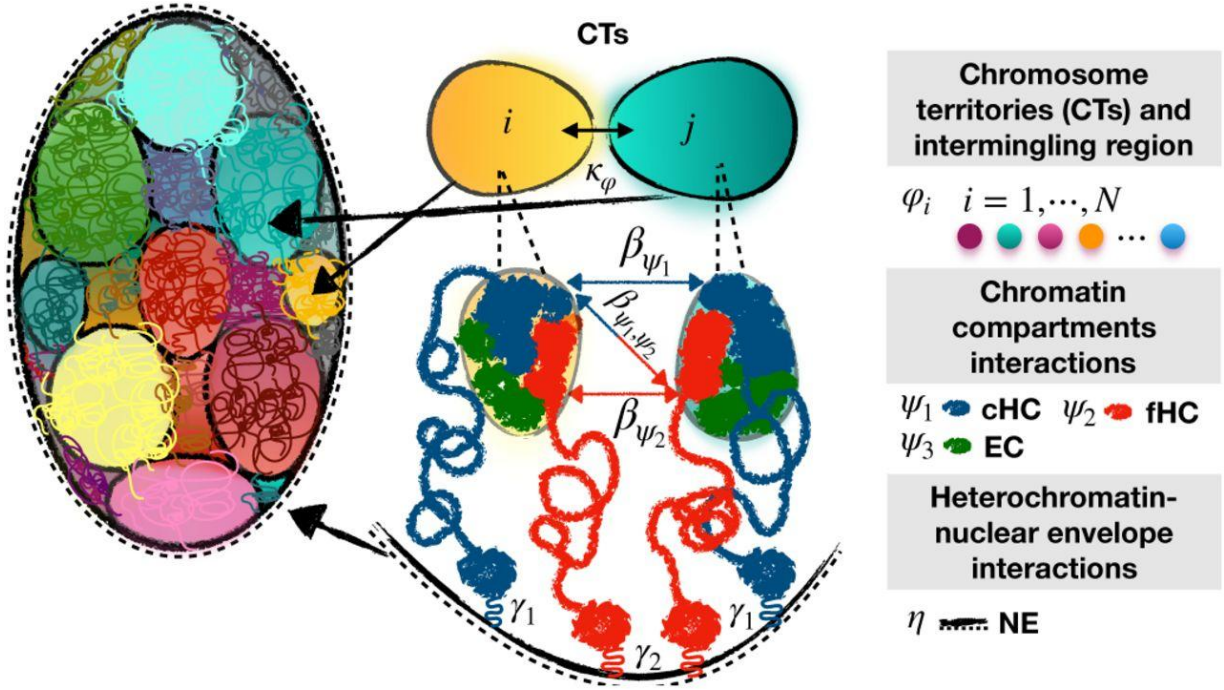
Figure 2.4 (a) A•T and (b) G•C base pairs, showing Watson–Crick hydrogen bonding.

Table 2.1 Hydrogen-Bond Distances (in Å) in Watson–Crick Base Pairs (Seeman *et al.*, 1976; Rosenberg *et al.*, 1976), in the Crystalline State. Estimated Standard Deviations are in Parentheses

U•A:	N3-H.....N1	2.835(8)
	O4.....H-N6	2.940(8)
C•G:	O2.....H-N2	2.86(1)
	N3.....H-N1	2.95(1)
	N4-H.....O6	2.91(1)

Levels of DNA organization

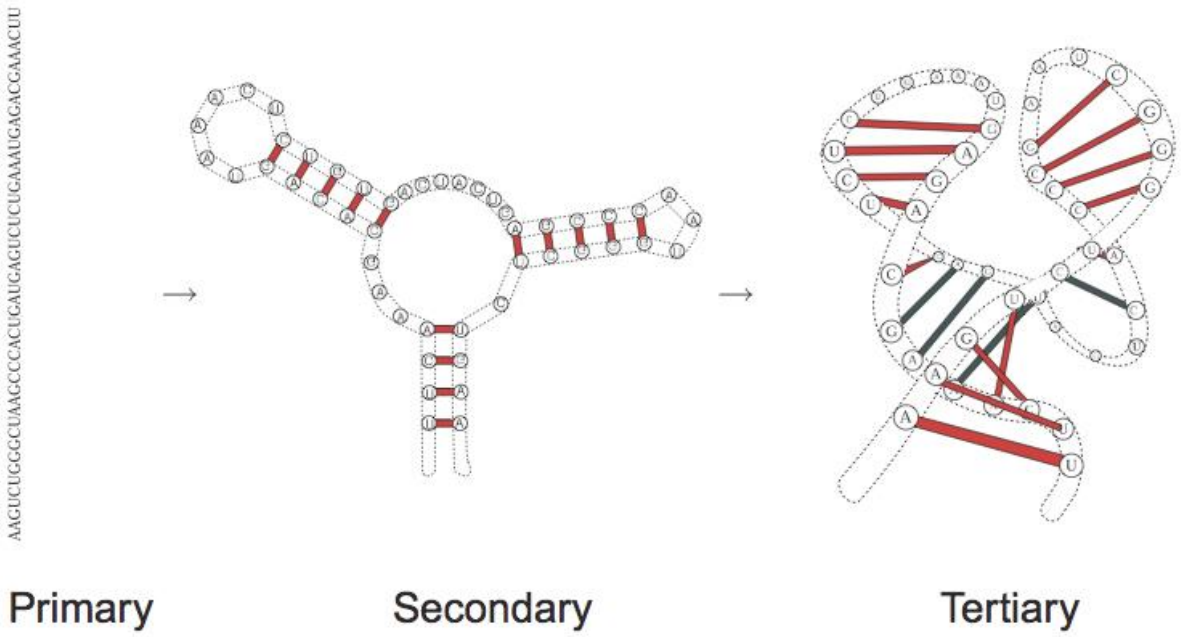




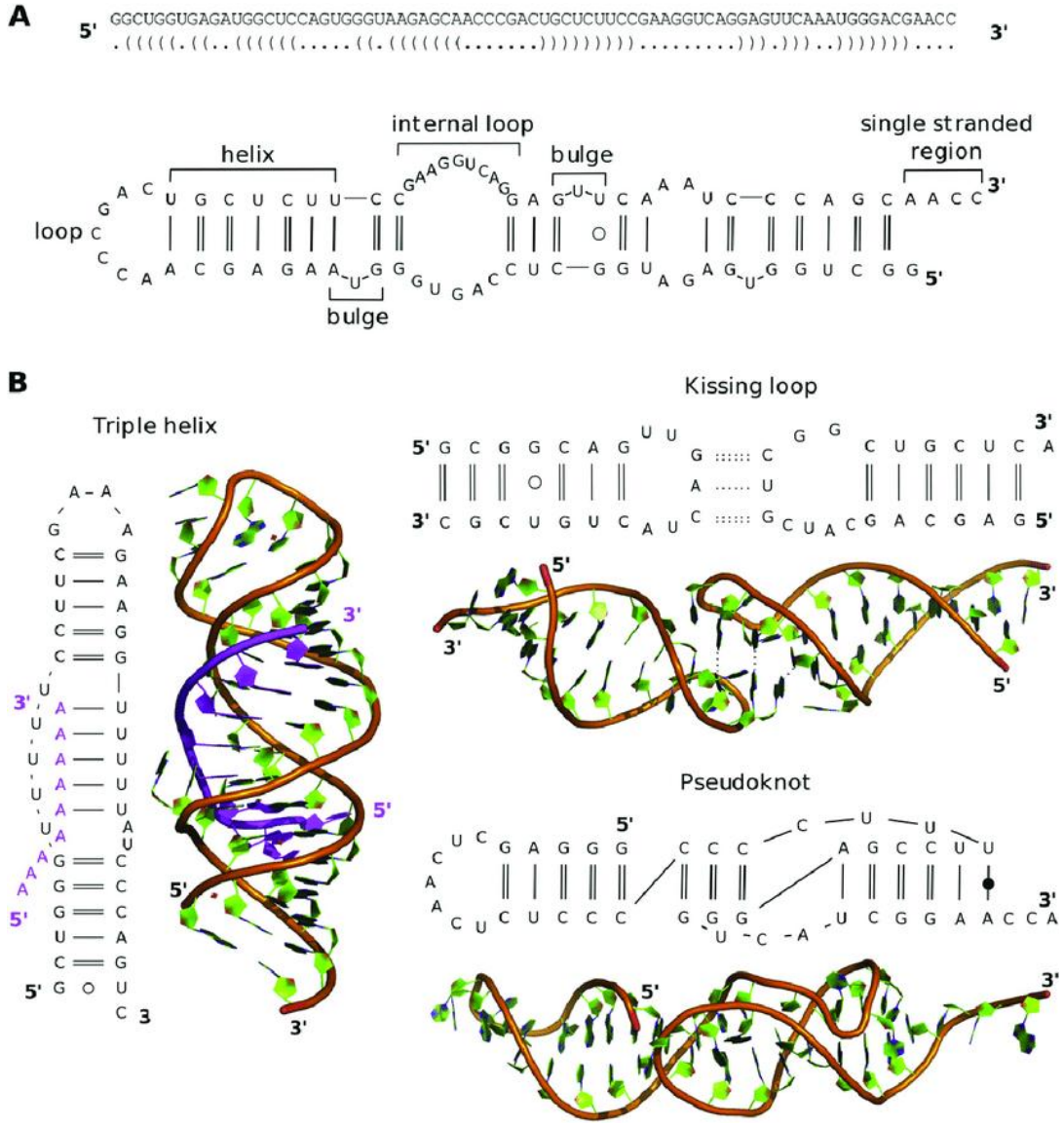
Schematic representation of the mulitphase liquid chromatin model of nucleus.

Rabia Laghmach, Michele Di Pierro, Davit A Potoyan, Biophysical Journal

RNA secondary and tertiary structure

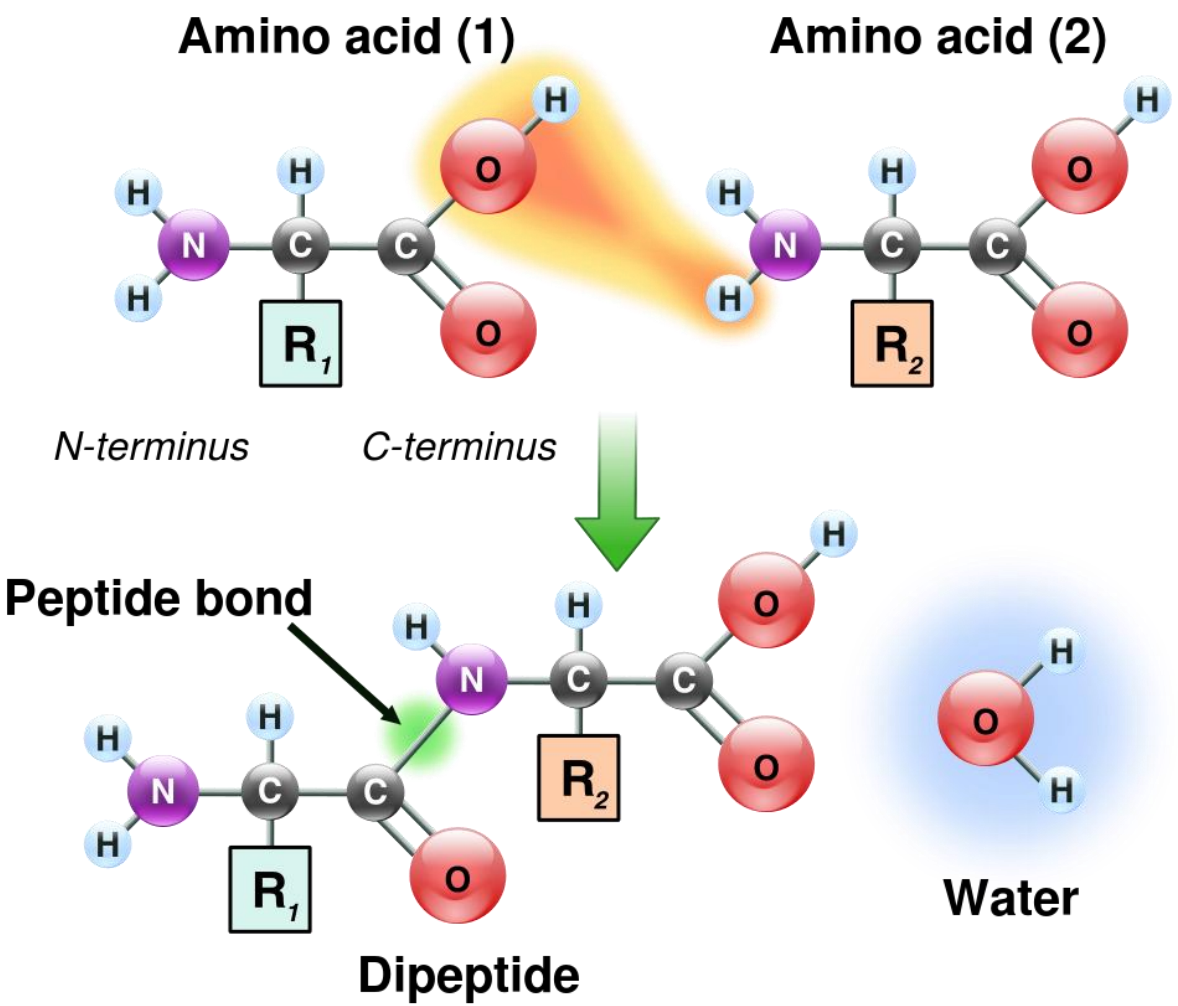


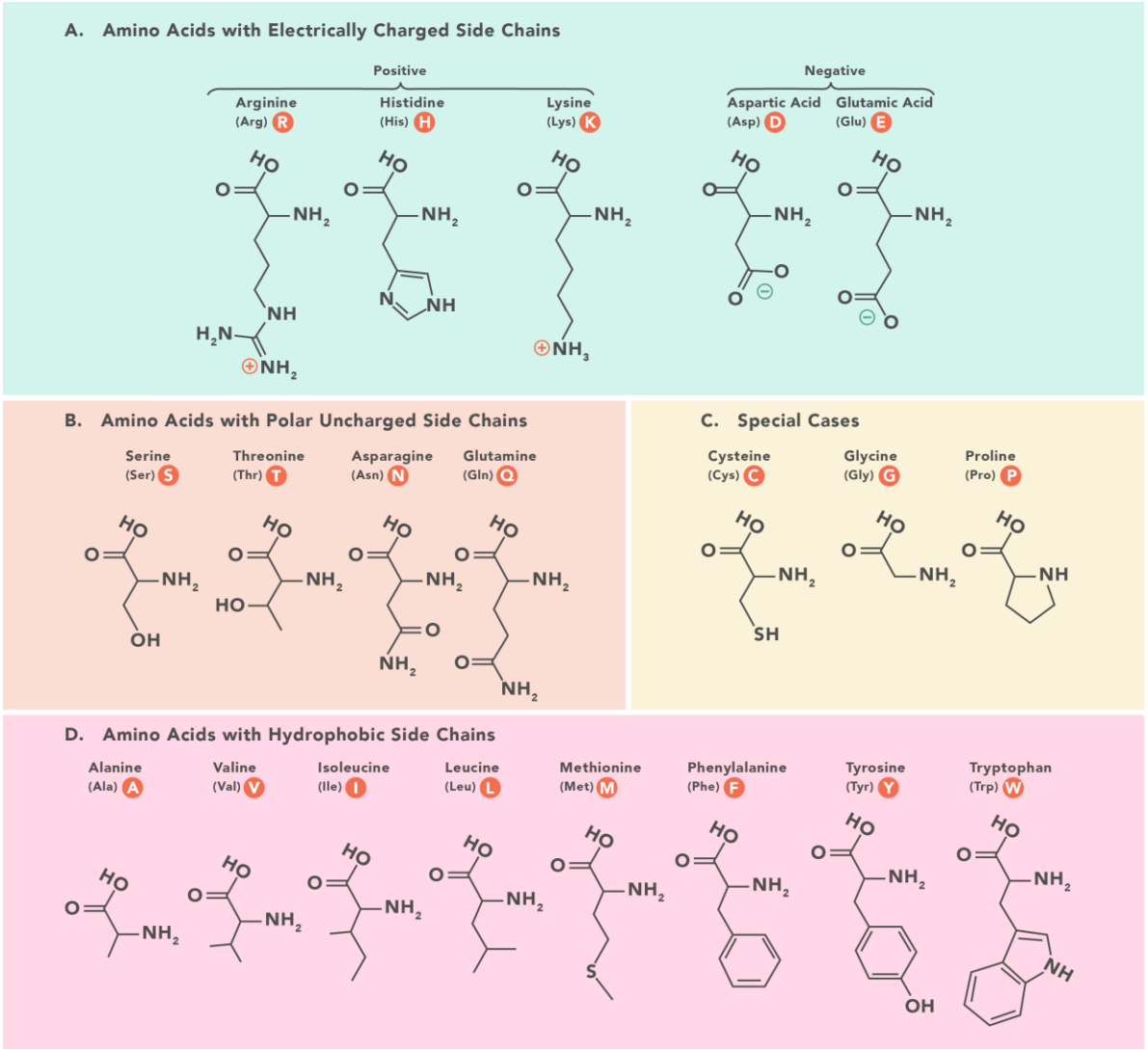
RNA is mostly a single strand molecule and can form nucleotide pairs with itself. Contrarily to DNA, RNA can also have structural roles.



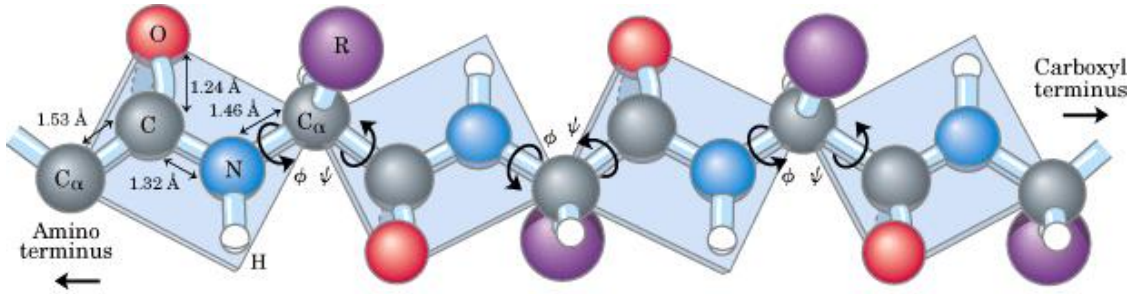
Common secondary and tertiary structures of RNA. A. The dot-bracket notation and 2-D map of a small region from the 5′ end of B2-SINE lncRNA showing common secondary structure elements. B. 2-D and 3-D representations of RNA tertiary structures. (Amanda J Lewis et al. 2015)

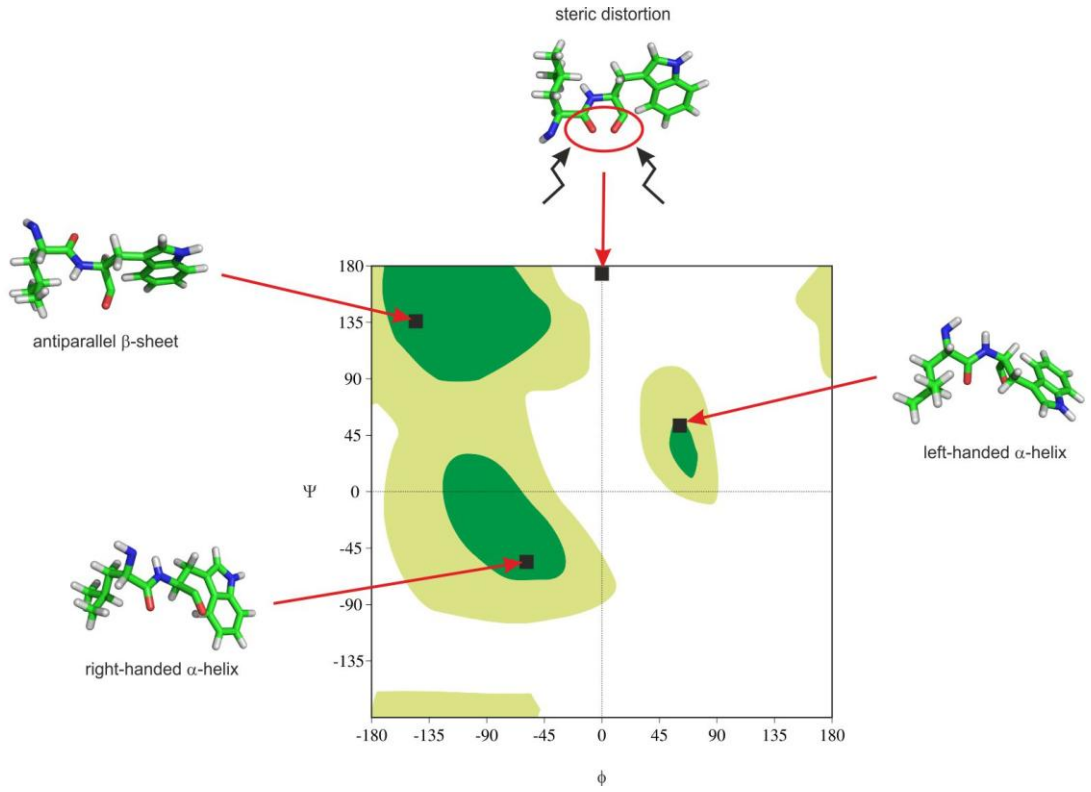
AMINO ACIDS AND PEPTIDE BONDS





Ramachandran Plots



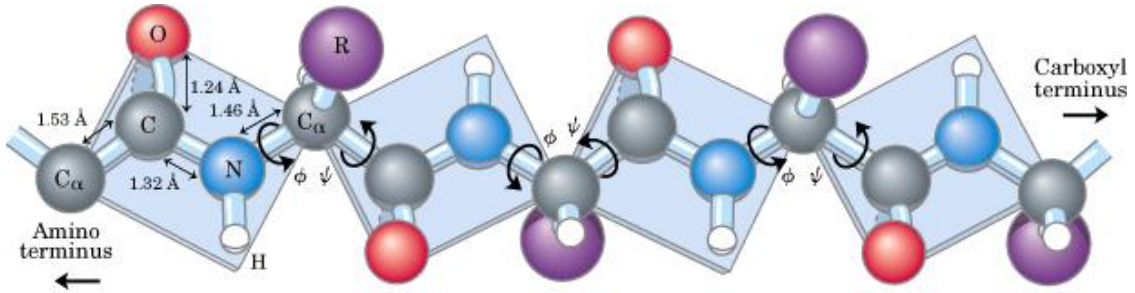


The relative position of two consecutive amino acids is described by two dihedral angles (φ and ψ).

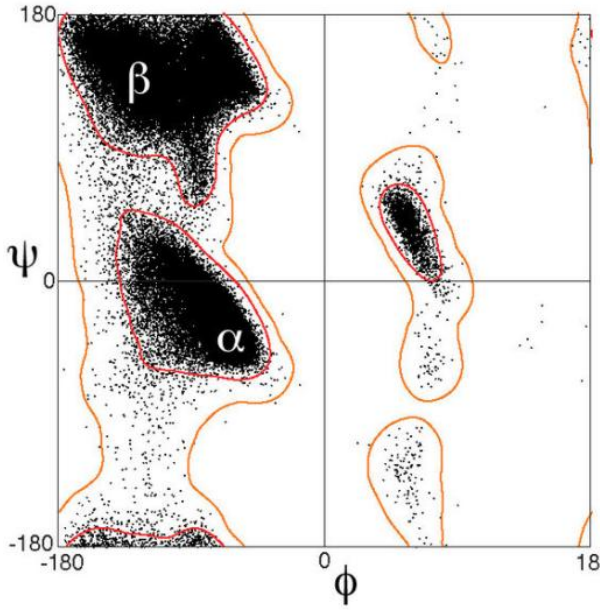
Not every combination of φ and ψ is possible, due to steric and electrostatic hindrance.

The values of the two dihedral angles is linked to the local secondary structure.

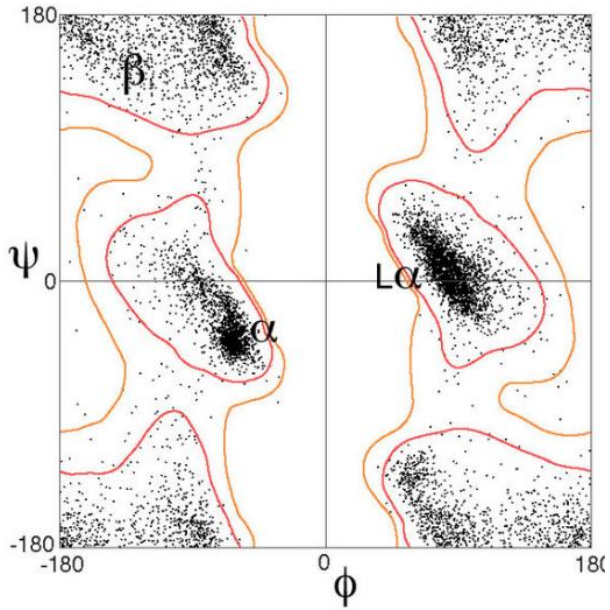
Ramachandran Plots



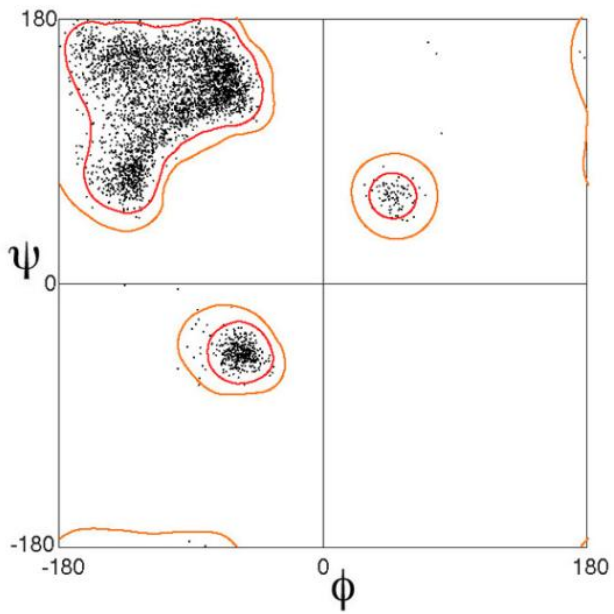
Different amino acids have different preferred orientation (which explain their secondary structure propensity).



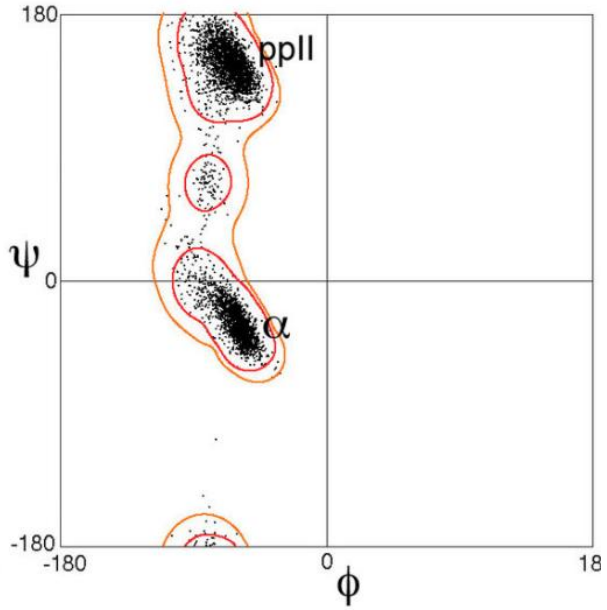
Average



Glycine



Proline



Pre-Proline

The Protein Data Bank and PDB files




```
HEADER    TRANSFERASE                             17-JUN-02   1M17
TITLE     EPIDERMAL GROWTH FACTOR RECEPTOR TYROSINE KINASE DOMAIN
TITLE    2 WITH 4-ANILINOQUINAZOLINE INHIBITOR ERLOTINIB
COMPND    MOL_ID: 1;
COMPND   2 MOLECULE: EPIDERMAL GROWTH FACTOR RECEPTOR;
COMPND   3 CHAIN: A;
COMPND   4 FRAGMENT: TYROSINE KINASE DOMAIN (RESIDUES 671-998);
COMPND   5 SYNONYM: RECEPTOR PROTEIN-TYROSINE KINASE ERBB-1;
COMPND   6 EC: 2.7.1.112;
COMPND   7 ENGINEERED: YES
SOURCE    MOL_ID: 1;
SOURCE   2 ORGANISM_SCIENTIFIC: HOMO SAPIENS;
SOURCE   3 ORGANISM_COMMON: HUMAN;
SOURCE   4 GENE: EGFR;
SOURCE   5 EXPRESSION_SYSTEM: SPODOPTERA FRUGIPERDA;
SOURCE   6 EXPRESSION_SYSTEM_COMMON: FALL ARMYWORM;
SOURCE   7 EXPRESSION_SYSTEM_STRAIN: AUTOGRAPHICA
SOURCE   8 CALIFORNICA/T.NICOPLUSIA;
SOURCE   9 EXPRESSION_SYSTEM_CELL_LINE: SF9;
SOURCE  10 EXPRESSION_SYSTEM_VECTOR_TYPE: PLASMID;
SOURCE  11 EXPRESSION_SYSTEM_PLASMID: PVL1392
KEYWDS    TRANSFERASE, TYROSINE KINASE DOMAIN
EXPDTA    X-RAY DIFFRACTION
AUTHOR    J.STAMOS,M.X.SLIWKOWSKI,C.EIGENBROT
REVDAT   2   25-FEB-03 1M17    1       JRNL
REVDAT   1   04-SEP-02 1M17    0
JRNL        AUTH   J.STAMOS,M.X.SLIWKOWSKI,C.EIGENBROT
JRNL        TITL   STRUCTURE OF THE EPIDERMAL GROWTH FACTOR RECEPTOR
JRNL        TITL 2 KINASE DOMAIN ALONE AND IN COMPLEX WITH A
JRNL        TITL 3 4-ANILINOQUINAZOLINE INHIBITOR.
JRNL        REF    J.BIOL.CHEM.                  V. 277 46265 2002
JRNL        REFN   ASTM JBCHA3  US ISSN 0021-9258
REMARK   1
REMARK   2
REMARK   2 RESOLUTION. 2.60 ANGSTROMS.
```

PDB: Protein Data Bank *https://www.rcsb.org/*

The Protein Data Bank and PDB files

Atom number, Atom "Type", Residue Name, Chain, Residue Number

Atom coordinates (Å)

Occupancy (fraction of time spent in that position for each atom)

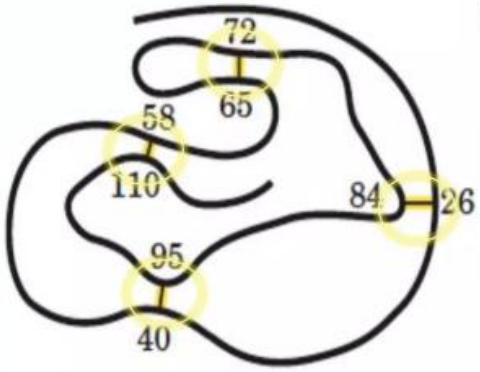
B-factors (fluctuations around the average positions Å^2)

Everything following the "ATOM" word indicates the properties of an atom in the protein

Element

```
ATOM      1  N   ARG A  20      24.557 -38.438  -2.219  1.00 48.18           N
ATOM      2  CA  ARG A  20      24.786 -38.383  -3.724  1.00 48.20           C
ATOM      3  C   ARG A  20      26.209 -38.780  -4.233  1.00 46.86           C
ATOM      4  O   ARG A  20      26.312 -39.492  -5.209  1.00 46.96           O
ATOM      5  CB  ARG A  20      24.328 -37.035  -4.295  1.00 48.44           C
ATOM      6  CG  ARG A  20      23.791 -37.076  -5.718  1.00 49.20           C
ATOM      7  CD  ARG A  20      23.543 -35.655  -6.309  1.00 49.60           C
ATOM      8  NE  ARG A  20      22.419 -34.974  -5.669  1.00 54.70           N
ATOM      9  CZ  ARG A  20      22.453 -33.754  -5.107  1.00 55.47           C
ATOM     10  NH1 ARG A  20      23.558 -33.001  -5.109  1.00 52.88           N
ATOM     11  NH2 ARG A  20      21.345 -33.268  -4.550  1.00 55.28           N
ATOM     12  N   SER A  21      27.281 -38.306  -3.583  1.00 45.68           N
ATOM     13  CA  SER A  21      28.663 -38.666  -3.946  1.00 43.64           C
ATOM     14  C   SER A  21      29.113 -39.741  -2.991  1.00 42.98           C
ATOM     15  O   SER A  21      28.807 -39.631  -1.810  1.00 43.65           O
ATOM     16  CB  SER A  21      29.618 -37.466  -3.846  1.00 43.49           C
ATOM     17  OG  SER A  21      29.413 -36.523  -4.890  1.00 41.49           O
ATOM     18  N   PHE A  22      29.822 -40.764  -3.483  1.00 41.35           N
ATOM     19  CA  PHE A  22      30.314 -41.886  -2.644  1.00 40.87           C
ATOM     20  C   PHE A  22      31.844 -42.145  -2.634  1.00 39.14           C
ATOM     21  O   PHE A  22      32.290 -43.097  -1.991  1.00 39.49           O
ATOM     22  CB  PHE A  22      29.550 -43.211  -2.938  1.00 41.87           C
ATOM     23  CG  PHE A  22      28.062 -43.084  -2.760  1.00 44.68           C
ATOM     24  CD1 PHE A  22      27.255 -42.664  -3.815  1.00 46.93           C
ATOM     25  CD2 PHE A  22      27.468 -43.322  -1.516  1.00 49.07           C
ATOM     26  CE1 PHE A  22      25.864 -42.500  -3.642  1.00 50.21           C
ATOM     27  CE2 PHE A  22      26.069 -43.156  -1.328  1.00 50.25           C
ATOM     28  CZ  PHE A  22      25.272 -42.759  -2.394  1.00 48.69           C
```

Anfisen's experiment

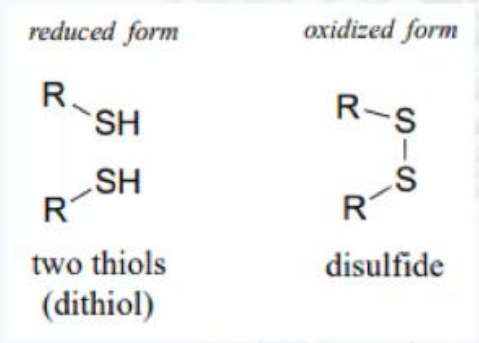


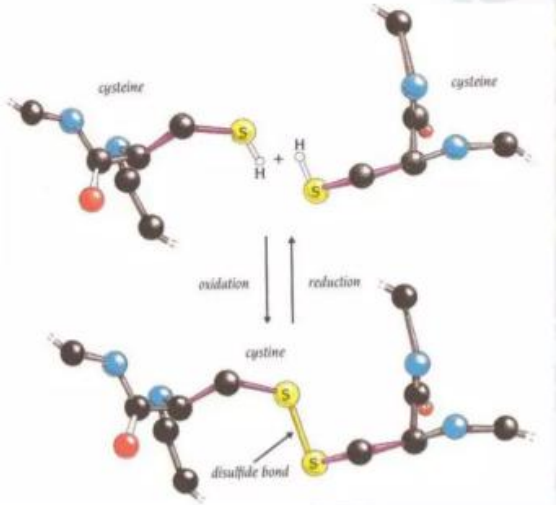
Native state of ribonuclease A

Native state; catalytically active.

Sulfhydryl group pairings in native conformation

58-110
72-65
26-84
40-95





Two cysteines in proximity can form a covalent bond called disulfide linkage.

SH groups are oxidized to form disulfide bonds.

2-MERCAPTOETHANOL

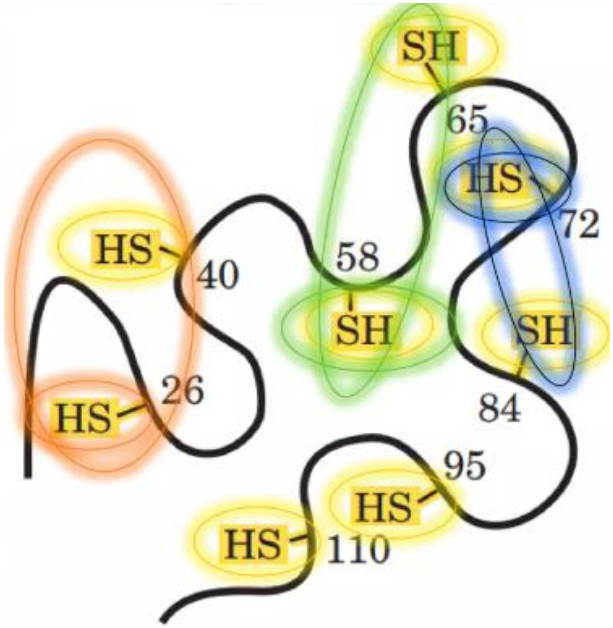
2-mercaptoethanol cleaves the disulfide bonds that may form between thiol groups of cysteine residues.

Urea

Urea interacts with the polar regions of the proteins resulting in the unfolding of the tertiary and secondary structure of the proteins and exposure of the hydrophobic core to water and urea, simply put- urea denatures proteins



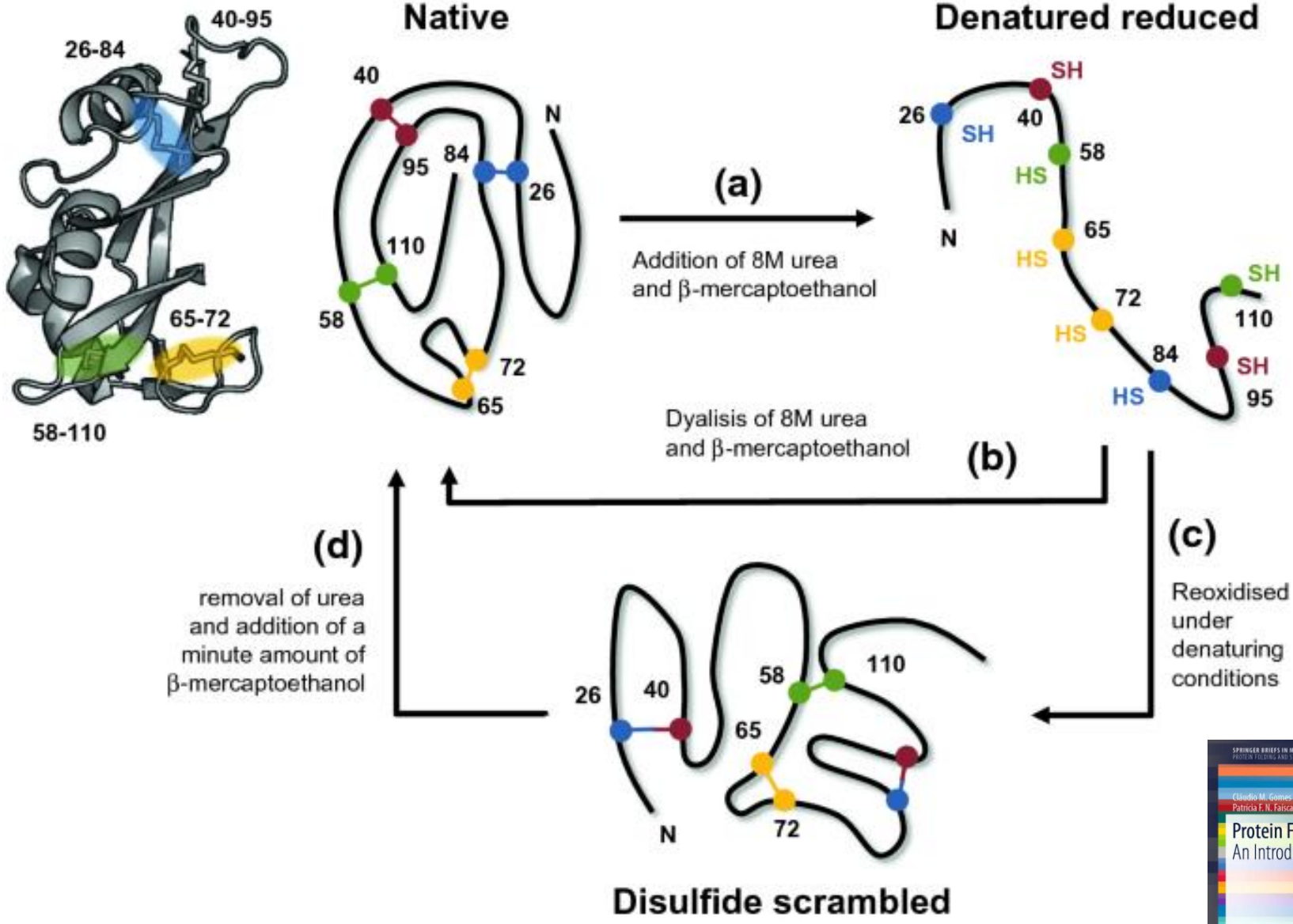
Anfisen's experiment

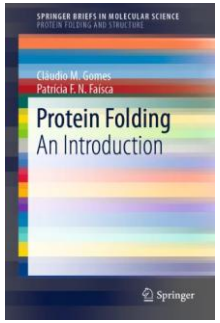


Lehninger Principles of Biochemistry, 8th Edition

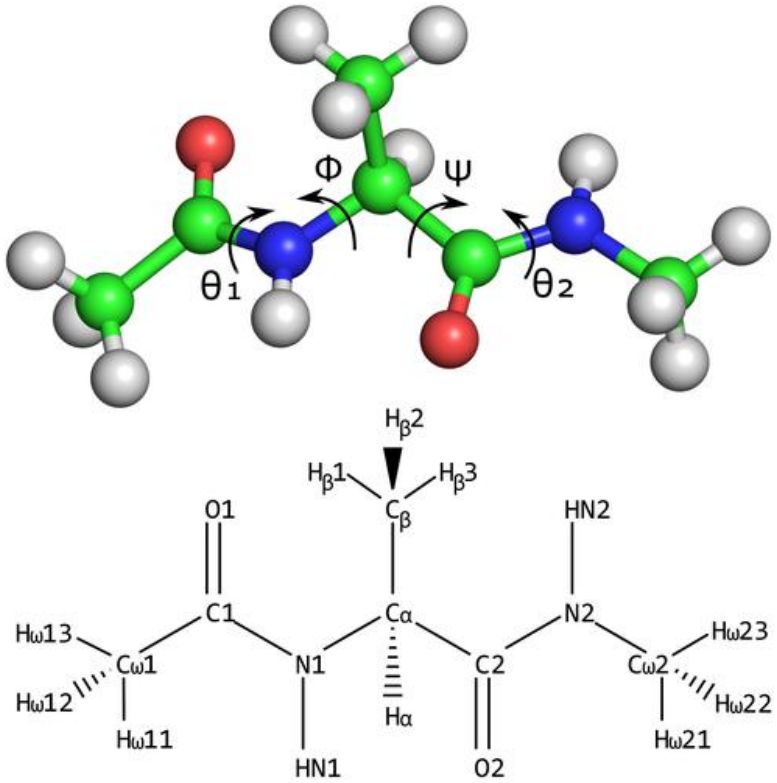
$7 \times 5 \times 3 \times 1 = 105$

Number of theoretical configuration in which ribonuclease can refold





Levinthal's paradox



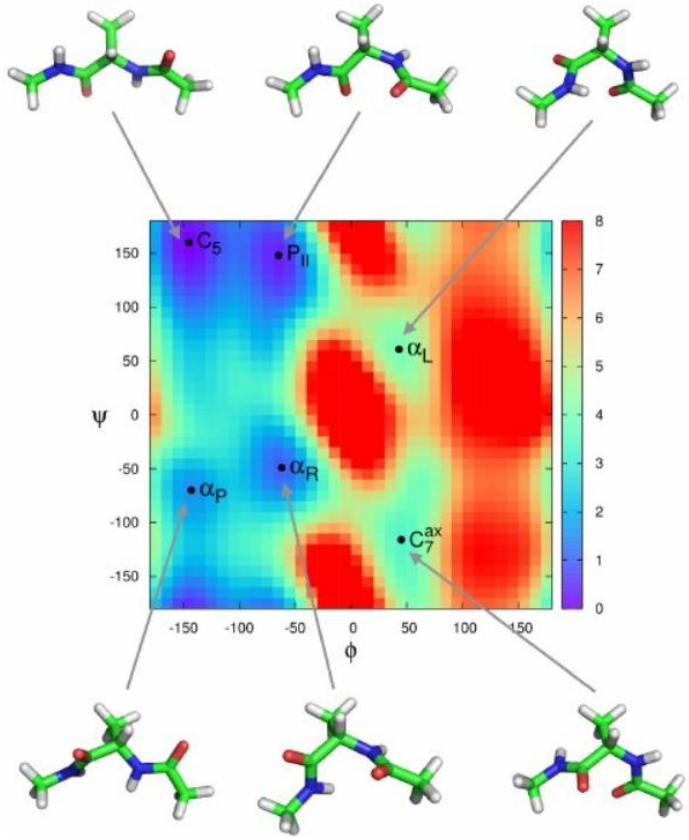


Figure 1: Ramachandran map of the alanine dipeptide in its (φ, ψ) space, with the locations of six local energy minima and their corresponding conformations (Chodera et al. 2006; Jaillet et al. 2011).

Protein length: 100 a.a.

Number of Ramachandran angles: 200

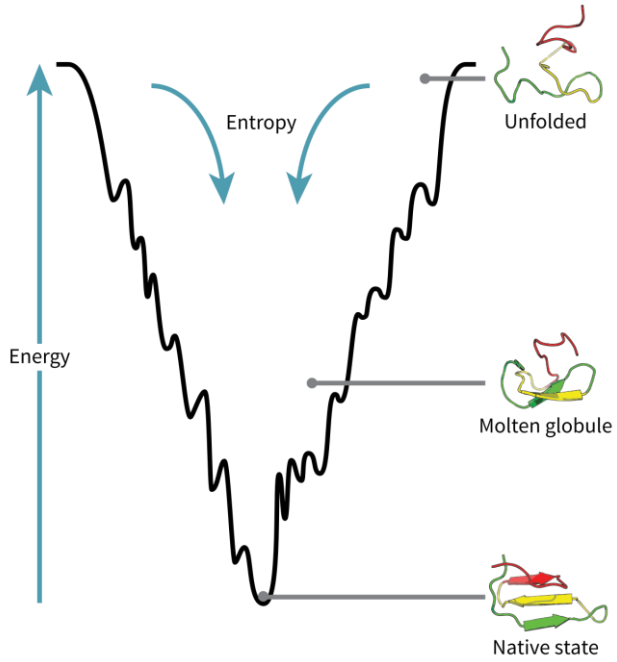
Possible state per each angle (average): 3

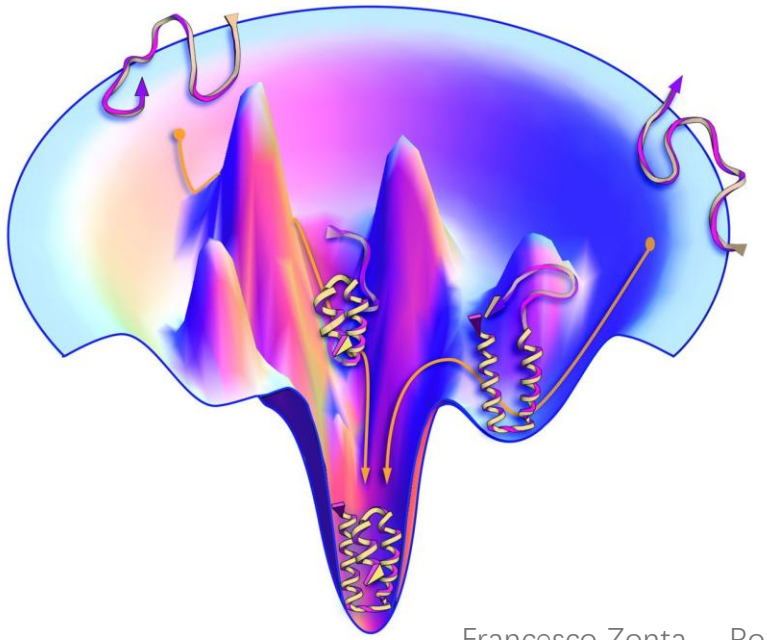
Number of possible combinations: $3^{200} = 2.7 \cdot 10^{95}$

Age of the universe (in seconds) : $4.35 \cdot 10^{17}$

Levinthal's paradox: Protein cannot explore the whole configuration space to find the native state (minimum of energy).

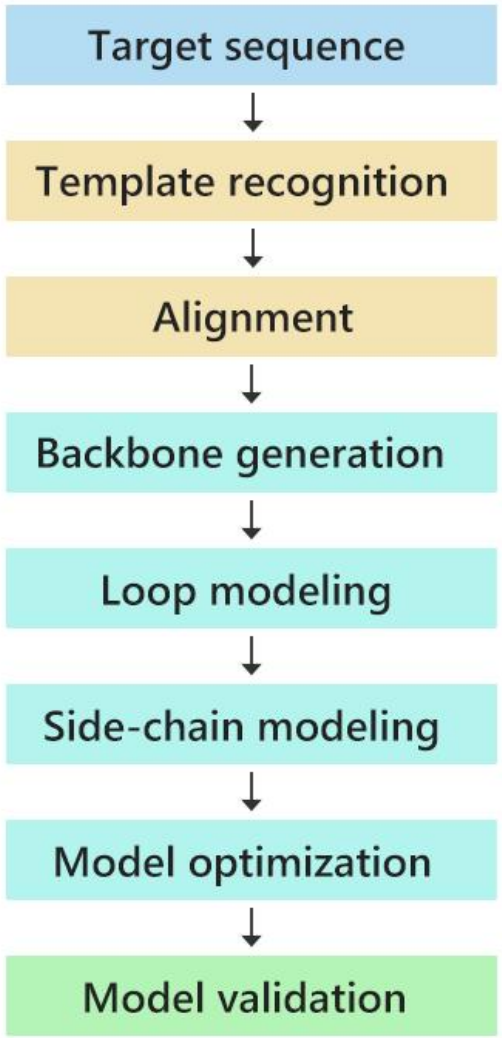
The protein folding funnel

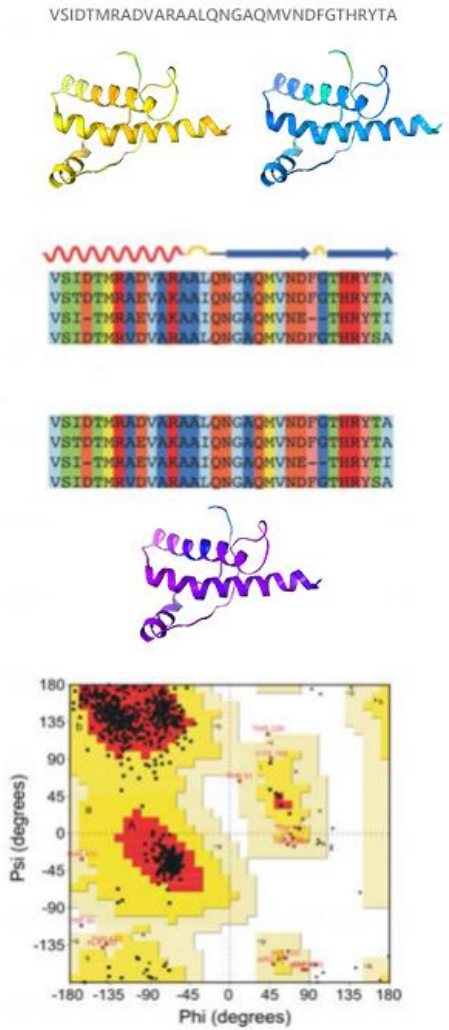




- A possible way to solve Levinthal's paradox is assuming that the energy landscape is shaped as a funnel.
- The funnel shape represents a bias toward the native state, with energy decreasing as the protein approaches its correctly folded structure.
- Multiple folding pathways can lead to the same native state, eliminating the need to search through all possible conformations randomly.
- The width of the funnel represents conformational entropy (number of possible states), which decreases as folding progresses and the structure becomes more ordered.
- Local minima in the funnel can act as folding intermediates or misfolded traps, but the overall downward slope drives folding toward the global energy minimum.
- Folding is primarily driven by the hydrophobic effect, where hydrophobic residues collapse into the protein core to minimize unfavorable contacts with water, while hydrophilic residues remain on the surface.

Homology Modeling

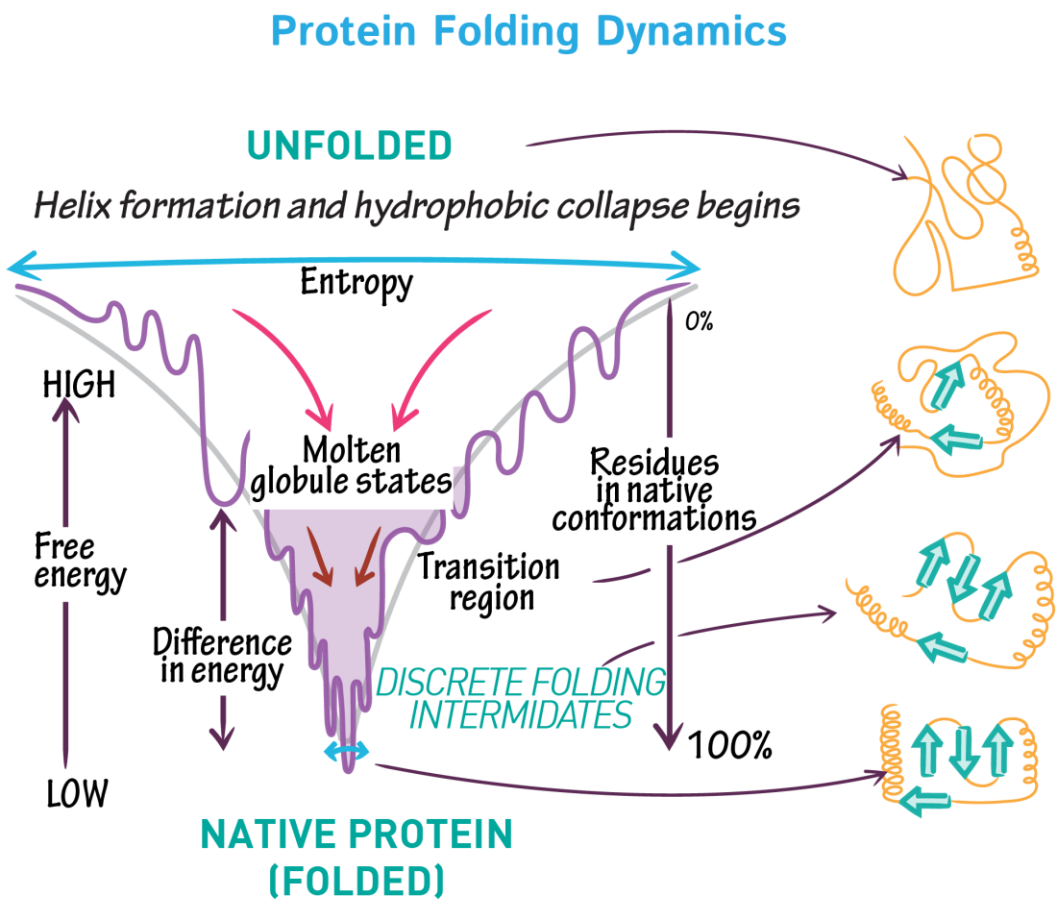




Homology modeling, also known as comparative modeling, uses homologous sequences with known 3D structures for the modeling and prediction of the structure of a target sequence

- Basic assumption: two homologous proteins will share very similar structures
- Although the number of actual proteins is vast, there is a limited set of tertiary structural motifs to which most proteins belong
- Homology modeling is one of the best-performing methods and (in most cases) gives accurate predicted models

Ab initio modeling



Ab initio: (Latin) from the beginning.

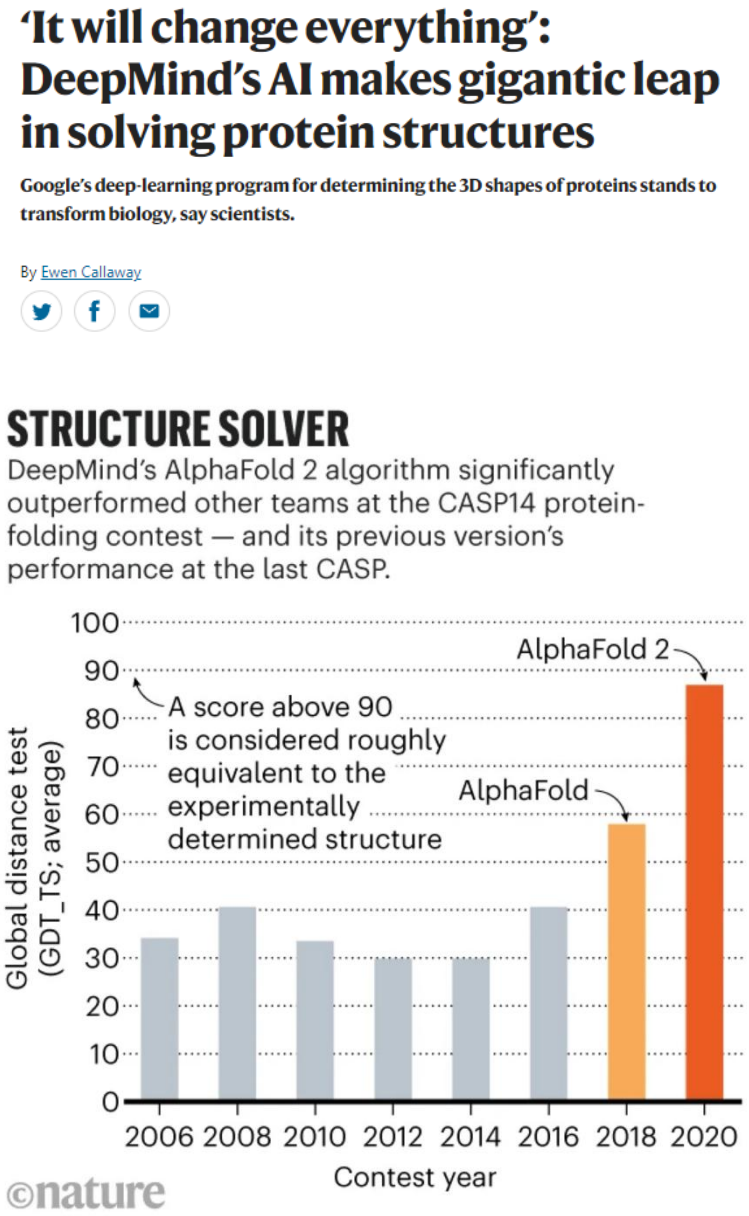
Ab initio methods aim to predict protein structures from their amino acid sequences without relying on known structural templates.

These methods often use physics-based energy functions or machine learning techniques to sample the conformational space and identify low-energy structures.

These methods are very time-consuming since the configuration space is incredibly large.

Very interesting from a theoretical point of view, but do not produce reliable results yet.

AlphaFold



AlphaFold utilizes deep learning and advanced machine learning techniques to accurately predict protein structures.

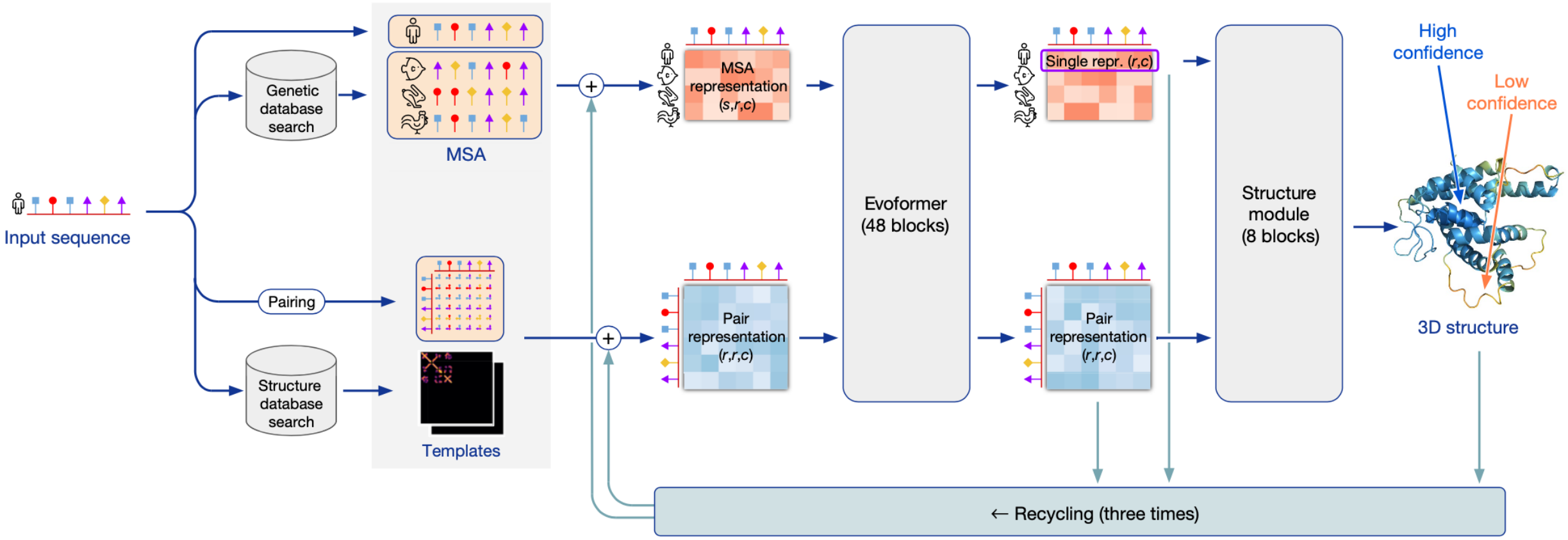
The key innovations of AlphaFold include:

1. Attention-based Neural Network: AlphaFold uses an attention-based neural network architecture that can capture long-range interactions between amino acids, which are crucial for accurate protein structure prediction.

2. **Evolutionary Information**: AlphaFold incorporates evolutionary information from multiple sequence alignments, which helps to improve the accuracy of its predictions.

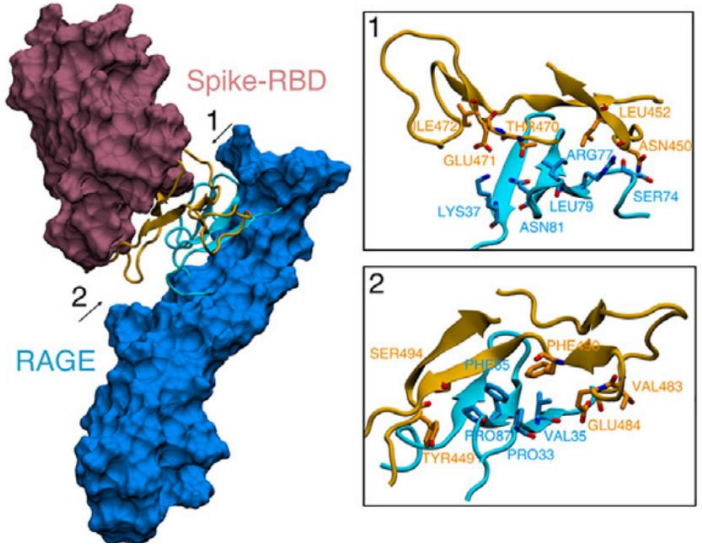
3. **End-to-End Approach:** AlphaFold is an end-to-end approach that directly predicts the three-dimensional structure from the amino acid sequence without relying on intermediate steps or templates.

How does it work?



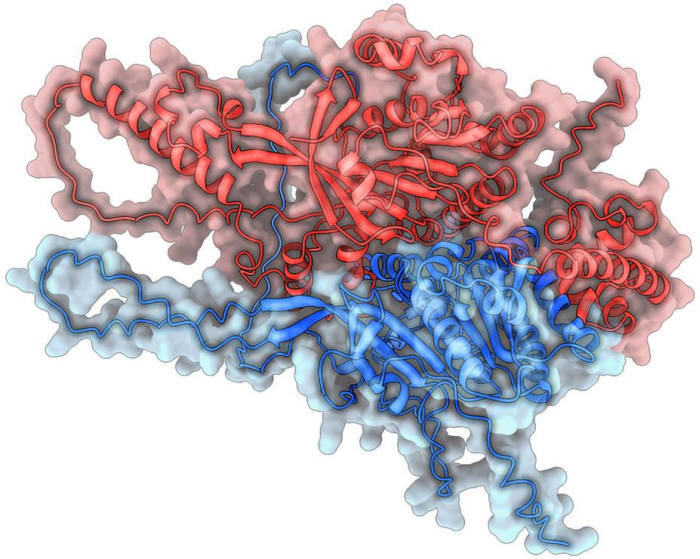
This and the next few slides are taken by a PPT from Kresten Lindorff-Larsen.

Molecular Docking



Protein-Ligand interactions

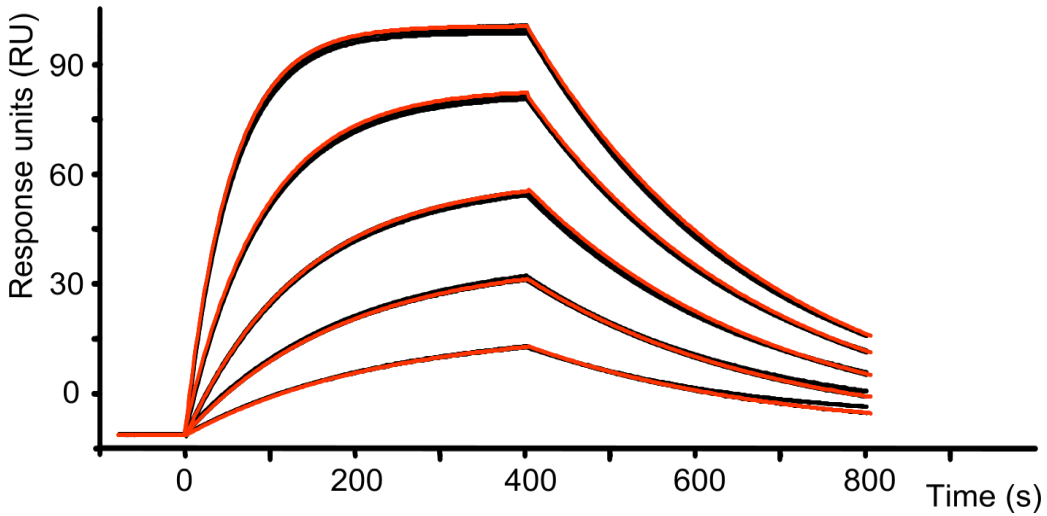
- Ligands are usually small and more rigid than proteins (few degrees of freedom).
- Small molecules usually bind to pockets (cavities).
- Interactions are often mediated by hydrophobic contributions.
- In many practical cases, a large number of ligands need to be tested (screened).



Protein-Protein interactions

- Proteins are large and can have important configuration changes upon binding.
- The interaction between the proteins happens on their surface.
- It is often mediated by hydrophilic residues and by surface complementarity.
- The interacting proteins are usually known (no need for screening).

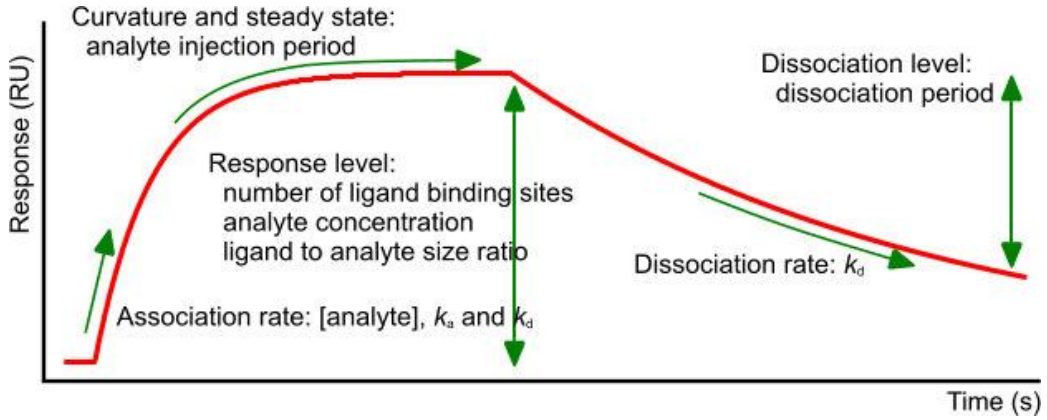
Molecular binding is a dynamical process

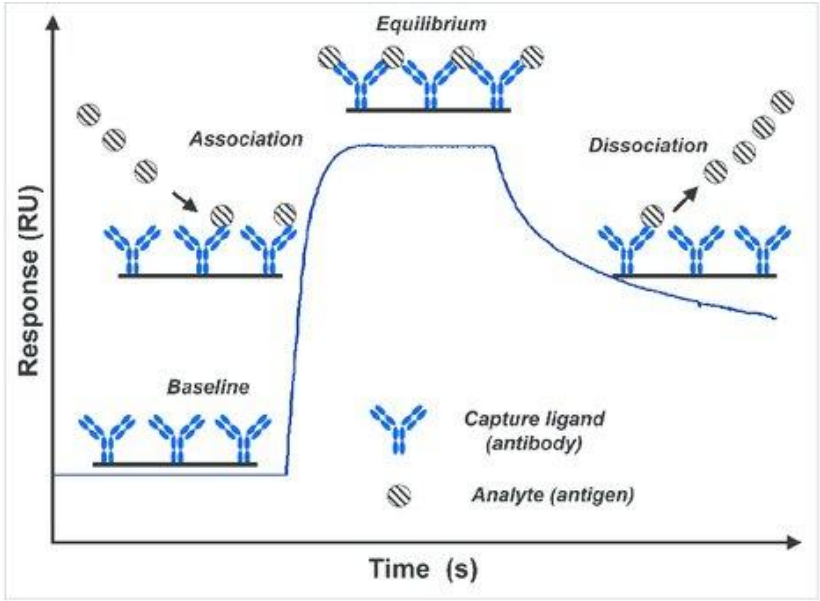


Typical SPR (Surface Plasmon Resonance) experiment.

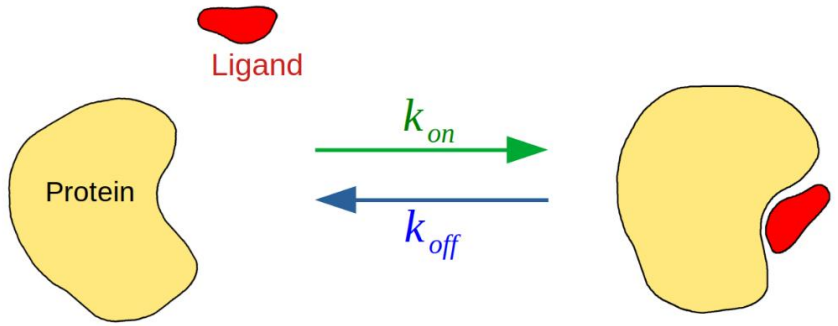
SPR measure binding and unbinding of two molecules as a function of the concentration.

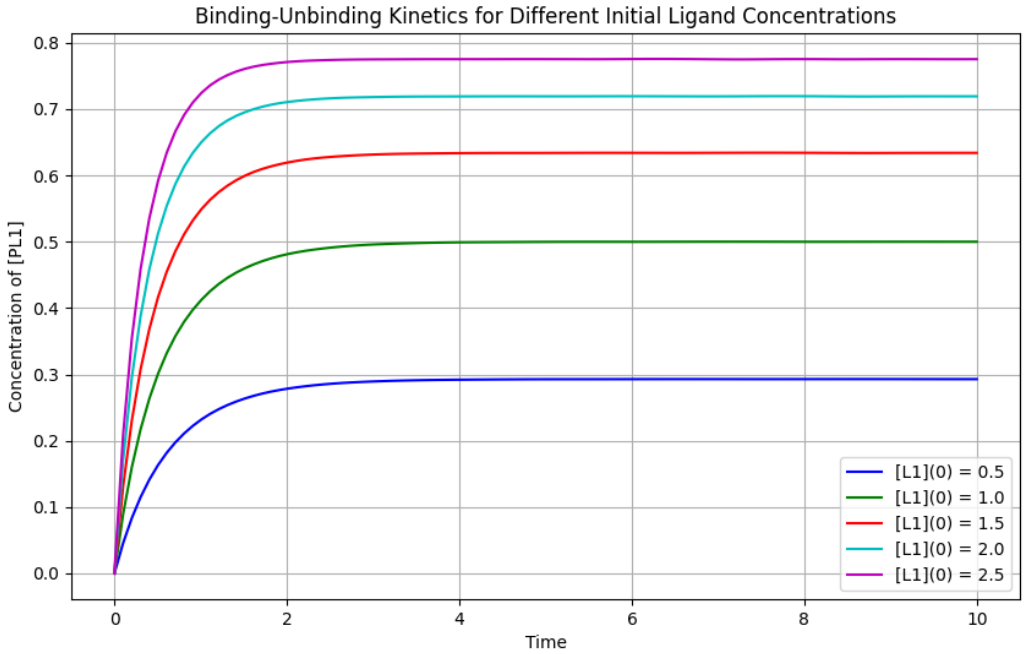
The ratio between the K_{on} and K_{off} is the equilibrium constant (K_D).





Molecular binding is a dynamical process





Typical SPR (Surface Plasom Resonance) experiment.

SPR measure binding and unbinding of two molecules as a function of the concentration.

The ratio between the K_{on} and K_{off} is the equilibrium constant (K_D).

Microscopically, binding works like a chemical reaction, with a dynamical equilibrium:

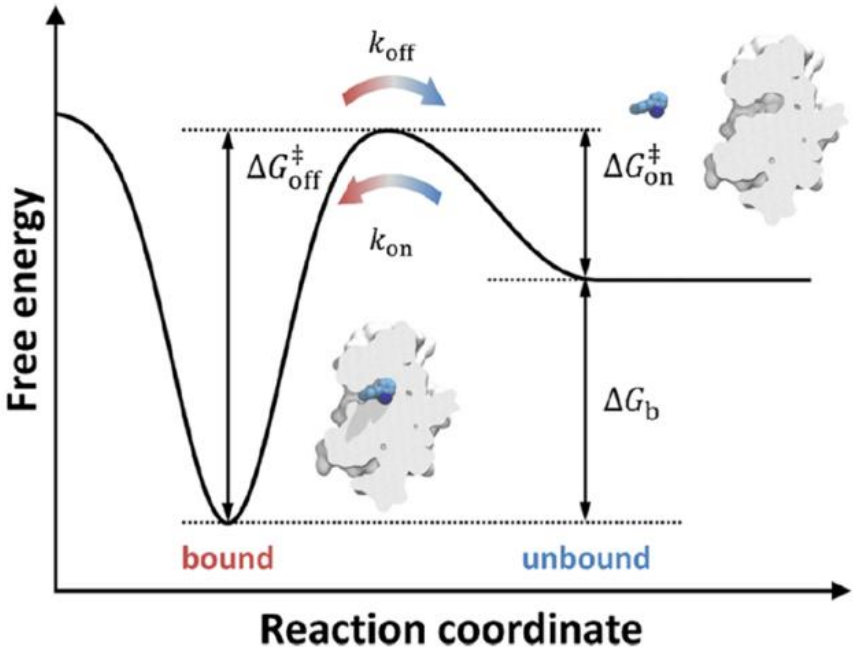
$$P + L_1 \leftrightharpoons PL_1$$

$$\frac{d[P]}{dt} = -k_{on}[P][L_1] + k_{off}[PL_1]$$

$$\frac{d[L_1]}{dt} = -k_{on}[P][L_1] + k_{off}[PL_1]$$

$$\frac{d[PL_1]}{dt} = k_{on}[P][L_1] - k_{off}[PL_1]$$

Binding Affinity and Binding Free Energy



$$K_D = e^{\Delta G/RT}$$

k_{on} and k_{off} are the rates at which one reaction and its reverse reaction happen.

The rates are proportional to the probability of "jumping" the free energy barriers, which is the exponential of the free energies:

$$k_{off} \propto e^{\Delta G^{\ddagger}_{off}/RT} \; ; \; k_{on} \propto e^{\Delta G^{\ddagger}_{on}/RT}$$

$$K_D = \frac{k_{off}}{k_{on}} = e^{(\Delta G^{\ddagger}_{off} - \Delta G^{\ddagger}_{on})/RT} = e^{\Delta G/RT}$$

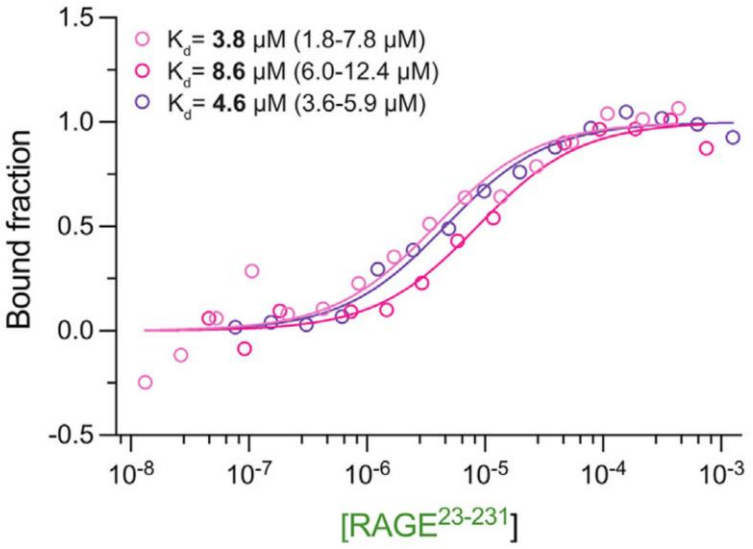
CHEMICAL REVIEWS
pubs.acs.org/CR Review

Thermodynamics and Kinetics of Drug-Target Binding by Molecular Simulation

Sergio Decherchi and Andrea Cavalli*

Cite This: https://dx.doi.org/10.1021/acs.chemrev.0c00534 Read Online

Binding Affinity and Binding Free Energy



The binding affinity K_A is the reciprocal of the dissociation constant:

$$K_A = \frac{1}{K_D}$$

Let us assume we have a generic chemical reaction: $A_xB_y \rightleftharpoons xA + yB$; we call *dissociation constant* the quantity:

$$K_D = \frac{[A]^x[B]^y}{[A_xB_y]}$$

If the reaction is 1:1 ($x = 1$, $y = 1$), the dissociation constant represents the concentration at which half of the molecules are interacting:

When $[B] = [AB]$, $K_D = [A]$: $\quad \frac{[AB]}{[B] + [AB]} = \frac{1}{2}$

The dissociation constant is linked to the binding free energy ΔG (difference of free energy between the bound and unbound state) by the following equation:

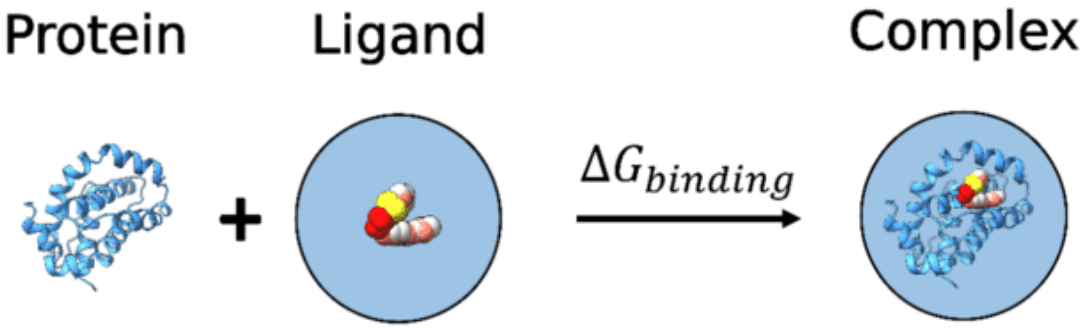
$$\Delta G = RTlnK_D \qquad K_D = e^{\Delta G/RT}$$

Energy measured in $KJ/mol \cdot K$

$$K_D = e^{\Delta G/K_BT}$$

Energy measured in K_BT

Binding Affinity and Binding Free Energy



ΔG is the difference of free energy between the bound and unbound state:

$$\Delta G = \Delta H - T\Delta S = G_B - G_U = U_B - U_U - TS_B - TS_U$$

$$\Delta H = \Delta U + p\Delta V = \Delta U$$ if no change in volume happens

ΔH : Energy changes associated with breaking and forming intermolecular interactions, (hydrogen bonds, van der Waals interactions, and electrostatic interactions), during the binding. Can be favorable or unfavorable.

Energy changes (usually unfavorable) due to the loss of solvation of the proteins and ligands.

ΔS : change in the configurational space accessible to the proteins and ligands upon binding (usually unfavorable).

Release or reorganization of water molecules from the binding interfaces (can contribute favorably)

Upon binding both enthalpy and entropy can decrease, but there are no general rule.

Binding Affinity and Binding Free Energy

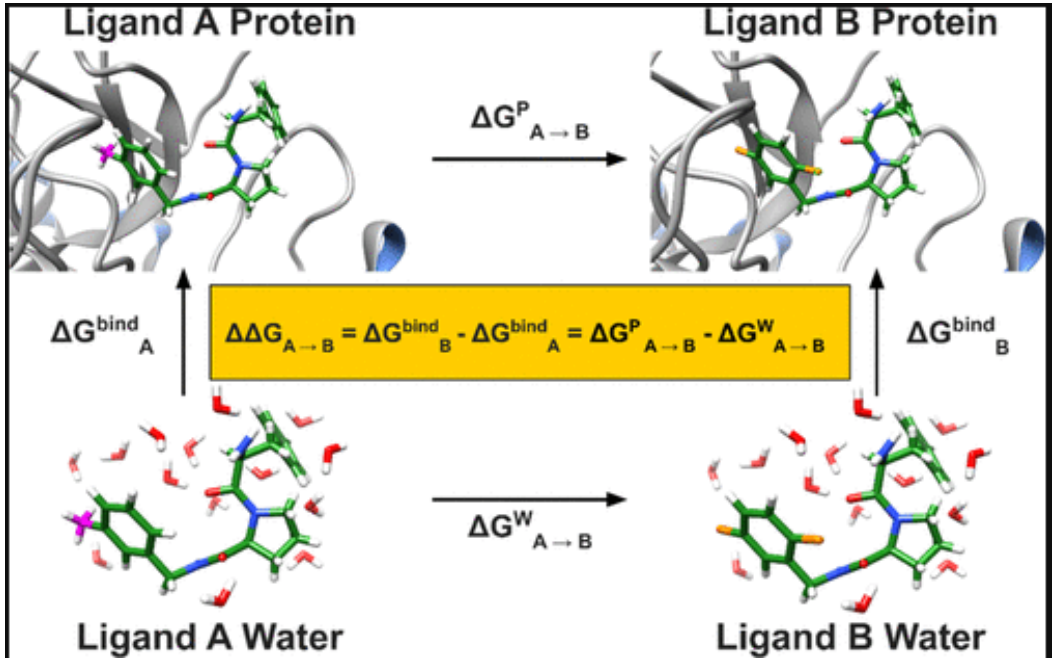
ΔG is the difference of free energy between the bound and unbound state:

$$\Delta G = \Delta H - T\Delta S = G_B - G_U = U_B - U_U - TS_B - TS_U$$

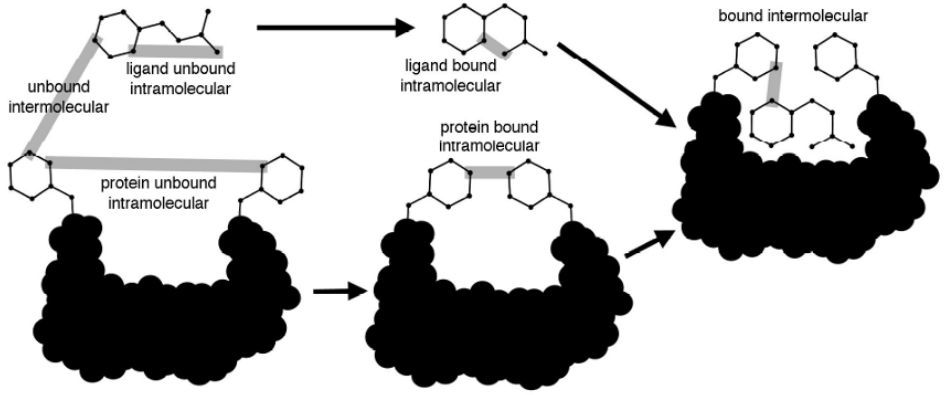
If we want to compare multiple ligands, we can calculate $\Delta\Delta G = \Delta G_B - \Delta G_A$, i.e. the difference between the two binding affinities

$$\frac{K_{DA}}{K_{DB}} = e^{(\Delta G_A - \Delta G_B)/K_B T} = e^{-\Delta\Delta G/K_B T}$$

If $\Delta\Delta G < 0$ B is a better binder than A
(lower binding energy, better binding affinity)



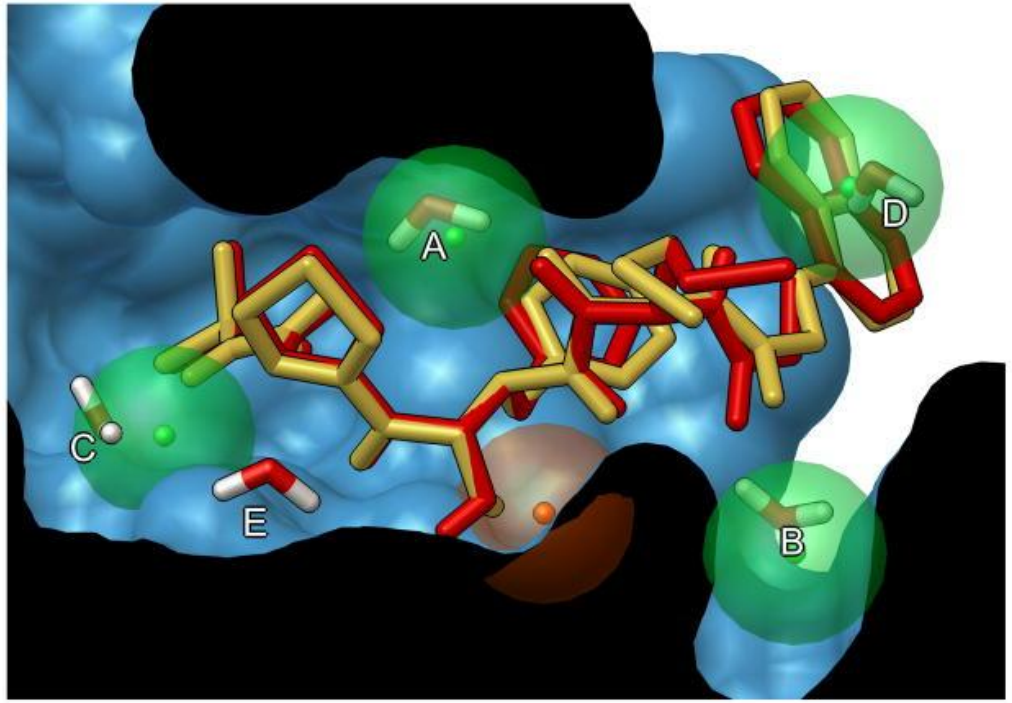
Molecular Docking: Autodock



The force field includes six pair-wise evaluations (V) and an estimate of the conformational entropy lost upon binding (ΔS_{conf}):

$$\Delta G = (V_{bound}^{L-L} - V_{unbound}^{L-L}) + (V_{bound}^{P-P} - V_{unbound}^{P-P}) + (V_{bound}^{P-L} - V_{unbound}^{P-L} + \Delta S_{conf})$$

where L refers to the "ligand" and P refers to the "protein" in a ligand-protein docking calculation.

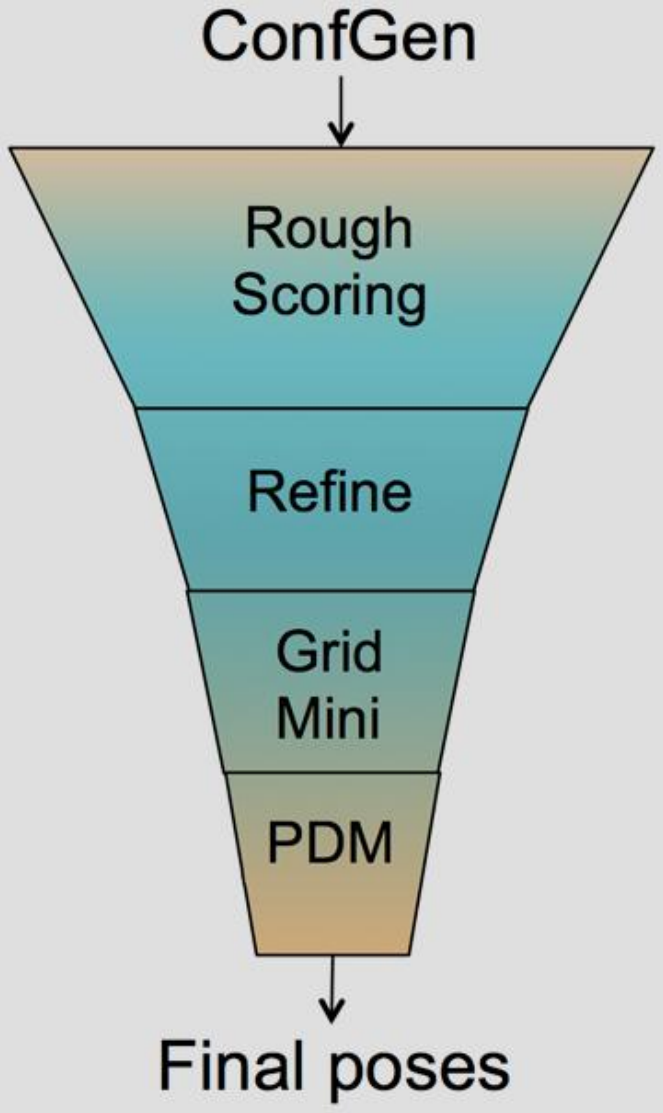


Each of the pair-wise energetic terms includes evaluations for dispersion/repulsion, hydrogen bonding, electrostatics, and desolvation:

$$V = W_{vdw}\sum_{i,j}\left(\frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^{6}}\right) + W_{hbond}\sum_{i,j}E(t)\left(\frac{C_{ij}}{r_{ij}^{12}} - \frac{D_{ij}}{r_{ij}^{10}}\right) + W_{elec}\sum_{i,j}\frac{q_i q_j}{e(r_{ij})r_{ij}} + W_{sol}\sum_{i,j}\left(S_i V_j + S_j V_i\right)e^{(-r_{ij}^2/2\sigma^2)}$$

Autodock potential is based on Physics, but it introduce weighting pre-factor to each term (W) to better fit empirical data.

Example: Schoedinger and Glide



Schoedinger is a software that use an empirical scoring function to determine the best mode and evaluate the binding energy of the docking.

The general form of the function is public, but the coefficients and the implementation are proprietary and not made public.

Schrödinger Company Platform Life Science Materials Science Pipeline

Overview Solutions Use Cases Resources Education

Home • Life science • Learn • Docking and scoring

$$\Delta G_{\text{bind}} = C_{\text{lipo-lipo}} \sum f(r_{\text{lr}}) + C_{\text{hbond-neut-neut}} \sum g(\Delta r)\, h(\Delta\alpha) + C_{\text{hbond-neut-charged}} \sum g(\Delta r)\, h(\Delta\alpha) + C_{\text{hbond-charged-charged}} \sum g(\Delta r)\, h(\Delta\alpha) + C_{\text{max-metal-ion}} \sum f(r_{\text{lm}}) + C_{\text{rotb}} H_{\text{rotb}} + C_{\text{polar-phob}} V_{\text{polar-phob}} + C_{\text{coul}} E_{\text{coul}} + C_{\text{vdW}} E_{\text{vdW}} + \text{solvation terms}$$

Friesner et al., Journal of Medicinal Chemistry 47:1739-49 (2004)

How do we use scoring functions

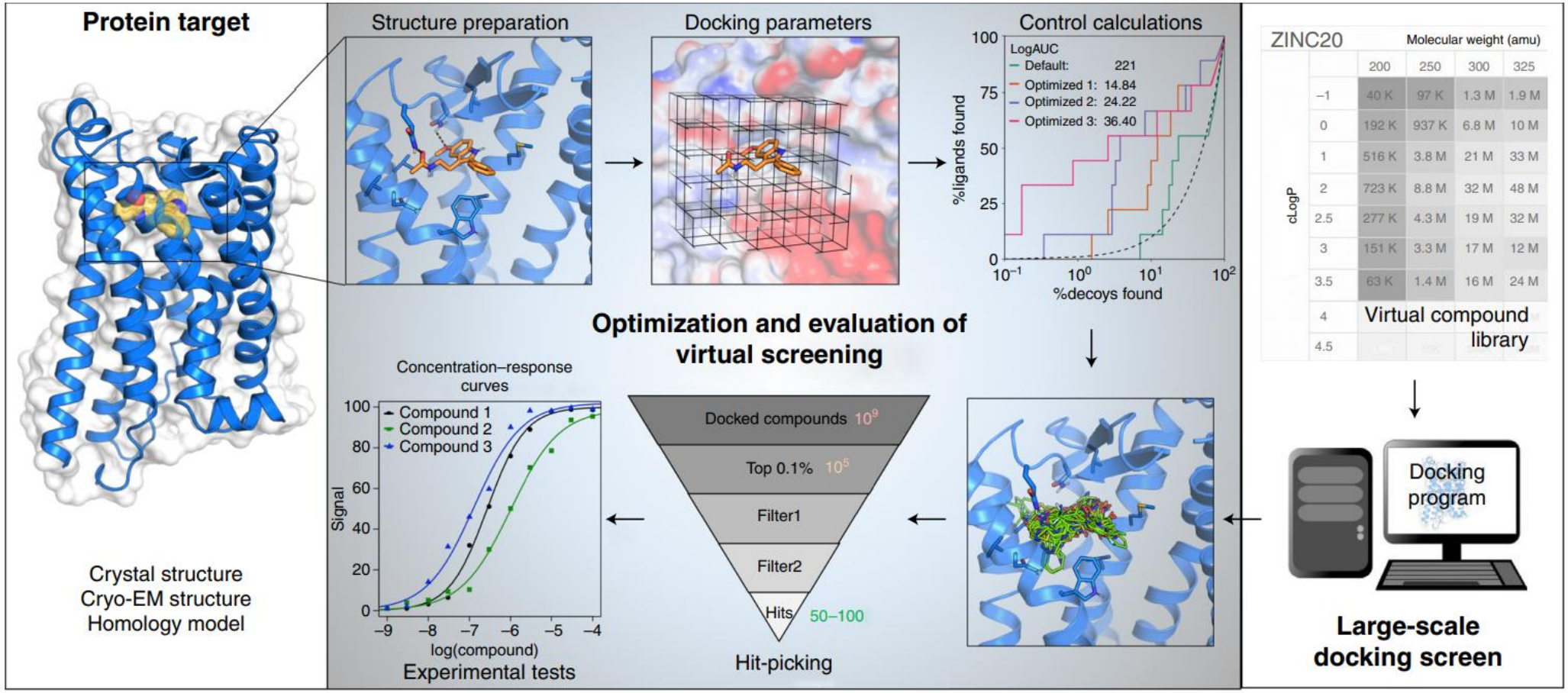
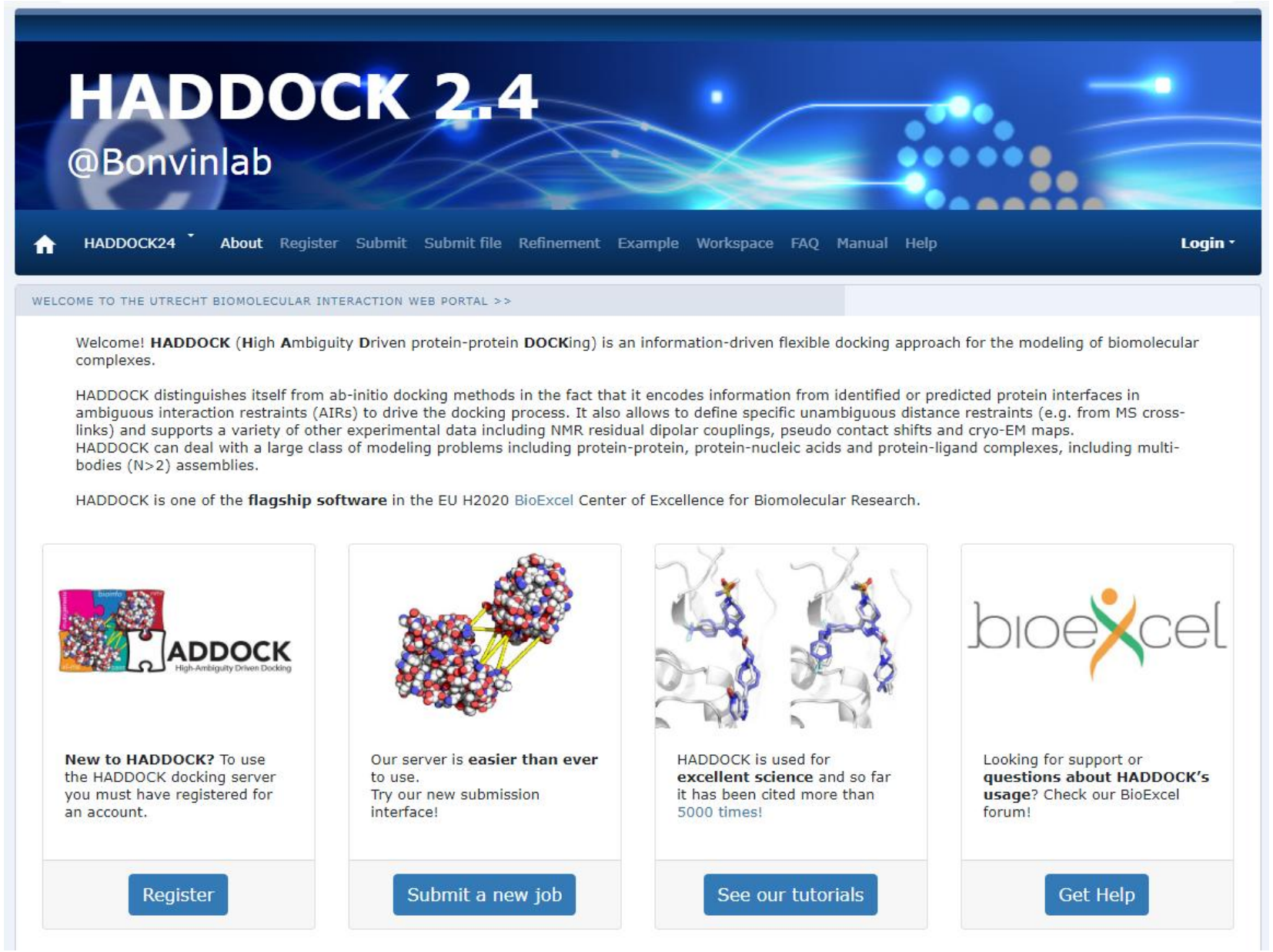


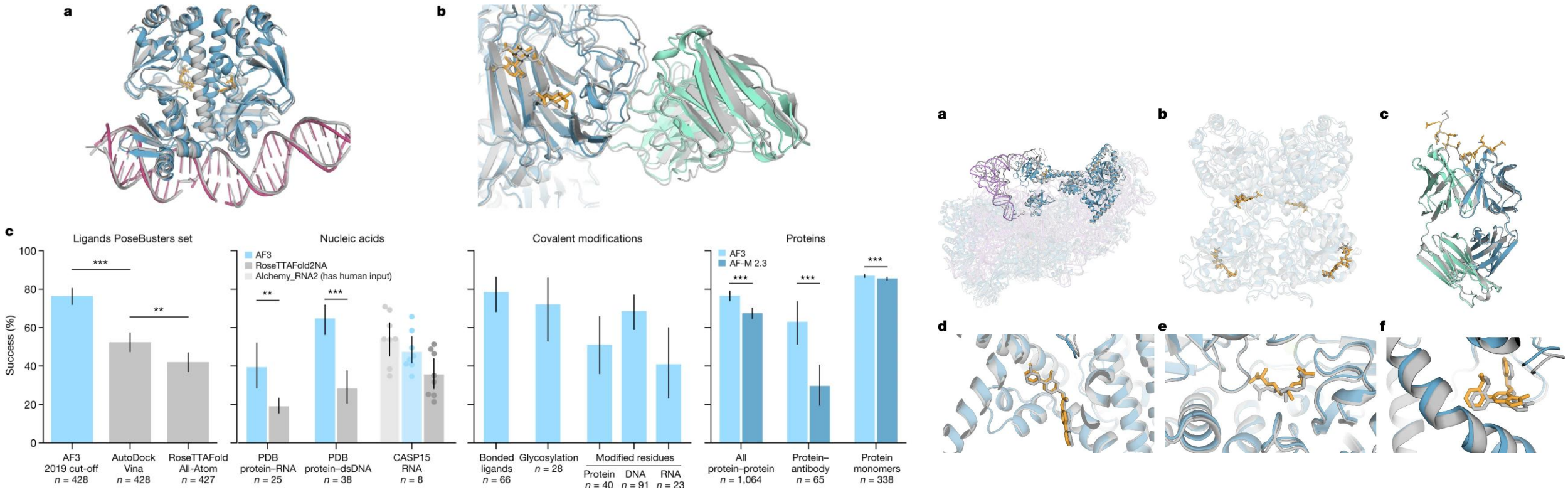
Fig. 1 | Large library docking workflow. The two required inputs for such a screen are the target structure and a screening database. Prior to using the database, the target structure must be converted into a representation used by the docking software and the pocket should be optimized with control calculations using retrospective analysis on known actives. After the prospective library has been docked, top-ranked hits can be filtered and selected for experiment. Multiple assays and controls are typically necessary to confirm activity.

Bender et. al. Nature Protocols 2021

Protein – Protein: Haddock



AF3 accurately predicts structures across biomolecular complexes



Selected structure predictions from AF3. Predicted protein chains are shown in blue (predicted antibody in green), predicted ligands and glycans in orange, predicted RNA in purple and the ground truth is shown in grey. **a**, Human 40S small ribosomal subunit (7,663 residues) including 18S ribosomal RNA and Met-tRNA$_i^{Met}$ (opaque purple) in a complex with translation initiation factors eIF1A and eIF5B (opaque blue; PDB 7TQL; full-complex LDDT, 87.7; GDT, 86.9). **b**, The glycosylated globular portion of an EXTL3 homodimer (PDB 7AU2; mean pocket-aligned r.m.s.d., 1.10 Å). **c**, Mesothelin C-terminal peptide bound to the monoclonal antibody 15B6 (PDB 7U8C; DockQ, 0.85). **d**, LGK974, a clinical-stage inhibitor, bound to PORCN in a complex with the WNT3A peptide (PDB 7URD; ligand r.m.s.d., 1.00 Å). **e**, (5S,6S)-O7-sulfo DADH bound to the AziU3/U2 complex with a novel fold (PDB 7WUX; ligand r.m.s.d., 1.92 Å). **f**, Analogue of NIH-12848 bound to an allosteric site of PI5P4Kγ (PDB 7QIE; ligand r.m.s.d., 0.37 Å).